United States Patent Application Publication	20250275853
Kind Code	A1
Publication Date	September 04, 2025
Inventor(s)	Nir; Noam et al.

LEAFLET CLAMPING MEMBERS

Abstract

The present disclosure relates to prosthetic valve that include leaflets coupled to a frame by clamping members, optionally in a sutureless manner. In an example, a prosthetic valve can include at least one stapling member coupling at least one leaflet of a plurality of leaflets to the frame, wherein the clamping member comprises a plurality of stapling portions separated from each other along a spine of the clamping member. Each stapling member can include a pair of legs extending from opposite ends of a backspan, wherein the legs extend through the leaflet and are bent around a corresponding strut of the frame, thereby coupling the leaflet to the frame.

Inventors: **Nir; Noam (Pardes-Hanna, IL), Bukin; Michael (Pardes Hanna, IL), Goldberg; Eran (Nesher, IL), Axelrod Manela; Noa (Kadima, IL), Sherman; Elena (Pardes Hana, IL)**

Applicant: **EDWARDS LIFESCIENCES CORPORATION** (Irvine, CA)

Family ID: **89542223**

Assignee: **EDWARDS LIFESCIENCES CORPORATION (Irvine, CA)**

Appl. No.: **19/212454**

Filed: **May 19, 2025**

Related U.S. Application Data

parent US continuation PCT/US2023/082574 20231205 PENDING child US 19212454
us-provisional-application US 63387909 20221216

Publication Classification

Int. Cl.: **A61F2/24** (20060101)

U.S. Cl.:

CPC **A61F2/2418** (20130101); A61F2220/0066 (20130101); A61F2240/001 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of PCT Patent Application No. PCT/US2023/082574, filed Dec. 5, 2023, which claims the benefit of U.S. Provisional Application No. 63/387,909, filed Dec. 16, 2022, each of which is incorporated by reference in its entirety herein.

FIELD

[0002] The present invention relates to prosthetic valves, and in particular, to prosthetic valves that include leaflets coupled to frames thereof by clamping members such as stapling members, compressible inserts, or gripping tubes.

BACKGROUND

[0003] Native heart valves, such as the aortic, pulmonary and mitral valves, function to assure adequate directional flow from, and to, the heart, and between the heart's chambers, to supply blood to the whole cardiovascular system. Various valvular diseases can render the valves ineffective and require replacement with artificial valves. Surgical procedures can be performed to repair or replace a heart valve. Conventional surgically implantable prosthetic valves typically include a leaflet assembly mounted within a relatively rigid support frame or ring. Components of the prosthetic valve are usually assembled with one or more biocompatible fabrics, and a fabric-covered sewing ring is provided around the valve for suturing to the tissue of the native leaflet.

[0004] Since surgeries are prone to an abundance of clinical complications, alternative less invasive techniques of delivering a prosthetic heart valve over a catheter and implanting it over the native malfunctioning valve have been developed over the years. Different types of prosthetic heart valves are known to date, including balloon expandable valve, self-expandable valves and mechanically-expandable valves. Different methods of delivery and implantation are also known, and may vary according to the site of implantation and the type of prosthetic valve. One exemplary technique includes utilization of a delivery assembly for delivering a prosthetic valve in a crimped state, from an incision which can be located at the patient's femoral or iliac artery, toward the native malfunctioning valve. Once the prosthetic valve is properly positioned at the desired site of implantation, it can be expanded against the surrounding anatomy, such as an annulus of a native valve, and the delivery assembly can be retrieved thereafter.

SUMMARY

[0005] Most expandable, prosthetic valves comprise an annular frame and prosthetic leaflets mounted inside the frame. The leaflets can be coupled along cusp edges thereof to the frame, either by being sutured to an inner skirt disposed around an inner surface of the frame, wherein the skirt is sutured, in turn, to the frame, or by the cusp edges being directly sutured to selected struts of the frame. The process of suturing leaflets to the frame, either directly to the struts or via an intermediate inner skirt member, can be relatively labor-intensive and involve relatively long process times, which can also result in increased costs. Accordingly, a need exists for improved leaflet attachment solutions to prosthetic valves, which can reduce process times, as well as the difficulty of the assembly procedure.

[0006] According to some aspects of the disclosure, there is provided a prosthetic valve comprising a frame movable between a radially compressed state and a radially expanded state, a valvular structure coupled to the frame and comprising a plurality of leaflets configured to regulate flow through the prosthetic valve, and at least one stapling member coupling at least one of the plurality of leaflets to the frame. The at least one stapling comprises a spine extending along a stapling member axis, and a plurality of stapling portions separated from each other by intermediate portions of the spine. Each stapling portion comprises at least one leg extending from a backspan. The at least one leg of each stapling portion extends through a corresponding one of the plurality of leaflets, and is bent around a corresponding strut of a plurality of struts of the frame, so as to couple the leaflet to the frame.

[0007] According to some aspects of the disclosure, there is provided a method of assembling a prosthetic valve comprising providing a stapling member in a first bent configuration, approximating a leaflet to a strut of a frame of a prosthetic valve, pushing the legs toward the leaflet such that tips of the legs pierce through the leaflet, extending the legs radially through the leaflet and across the strut, and transitioning the stapling member to a second bent configuration by bending the legs at second bends thereof over the struts, so as to couple the leaflet to the strut. The stapling member in a first bent configuration comprises a spine extending along a stapling member axis and a plurality of U-shaped stapling portions separated from each other by intermediate portions of the spine, each stapling portion comprising a backspan and a pair of legs bent relative to the backspan at first bends of the legs.

[0008] According to some aspects of the disclosure, there is provided a prosthetic valve comprising a frame movable between a radially compressed state and a radially expanded state, a valvular structure coupled to the frame and comprising a plurality of leaflets configured to regulate flow through the prosthetic valve, and at least one compressible insert disposed within a recess formed in at least one of the recessed struts. The frame comprises a plurality of intersecting struts, the plurality of intersecting struts comprising recessed struts and unrecessed struts. The at least one of the plurality of leaflets is coupled to at least one of the recessed struts by having a cusp end portion thereof clamped inside the recess of the corresponding recessed strut by the corresponding at least one compressible insert.

[0009] According to some aspects of the disclosure, there is provided a method of assembling a prosthetic valve comprising approximating a cusp end portion of at least one leaflet to an inner surface of a recessed strut of a frame of a prosthetic valve, and pushing a compressible insert, along with a portion of the cusp end portion, into a recess formed in the corresponding recessed strut, thereby coupling the cusp end portion to the frame.

[0010] According to some aspects of the disclosure, there is provided a prosthetic valve comprising a frame comprising a base portion and a plurality of vertically oriented commissure posts extending from the base portion, a valvular structure coupled to the frame and comprising a plurality of leaflets configured to regulate flow through the prosthetic valve, and at least one gripping tube. The at least one gripping tube comprises an inner channel and a slot extending between free ends of side arms of the gripping tube. The at least one of the plurality of leaflets is coupled to the frame by the at least one gripping tube disposed around a cusp end portion of the leaflet which is at least partially wrapped around at least a part of the base portion retained within the inner channel.

[0011] According to some aspects of the disclosure, there is provided a method of assembling a prosthetic valve comprising positioning a cusp end portion of at least one leaflet between a base portion of a frame of a prosthetic valve and at least one gripping tube, and pushing the gripping tube toward the cusp end portion and the base portion, such that a slot of the gripping tube expands to let the base portion and the cusp end portion slide into an inner channel of the gripping tube, until the base portion resides within the inner channel with the cusp end portion at least partially wrapped around the base portion, at which point side arms defining the slot of the gripping tube are allowed to snap back toward each and clamp against a portion of the leaflet extending through the slot, so as to couple the at least one leaflet to the base portion.

[0012] The aspects of this disclosure can be used in combination or separately. This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the detailed description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. The foregoing and other objects, features, and advantages of the invention will become more apparent from the following detailed description, which proceeds with reference to the accompanying figures.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0013] Some examples of the invention are described herein with reference to the accompanying figures. The description, together with the figures, makes apparent to a person having ordinary skill in the art how some examples may be practiced. The figures are for the purpose of illustrative description and no attempt is made to show structural details of an example in more detail than is necessary for a fundamental understanding of the invention. For the sake of clarity, some objects depicted in the figures are not to scale.

[0014] In the Figures:

[0015] FIG. **1**A is a perspective view of an exemplary prosthetic valve that includes a leaflet structure coupled to an inner skirt.

[0016] FIG. **1**B is a perspective view of the prosthetic valve of FIG. **1**A, without the outer skirt.

[0017] FIG. **1**C is a perspective view of a frame of the prosthetic valve of FIGS. **1**A-**1**B.

[0018] FIG. **2** shows leaflets of a valvular structure of the prosthetic valve of FIGS. **1**A-**1**B.

[0019] FIG. **3** is a perspective view of an exemplary prosthetic valve that includes a leaflet structure sutured to struts of the frame.

[0020] FIG. **4** shows an exemplary delivery assembly comprising a delivery apparatus carrying a prosthetic device.

[0021] FIG. **5**A shows a portion of an exemplary stapling member.

[0022] FIG. **5**B is a sectional view taken along line **5**B-**5**B of FIG. **5**A.

[0023] FIG. **5**C shows an enlarged view of an end portion of a leg of the stapling member of FIG. **5**A.

[0024] FIG. **6**A shows the stapling member of FIG. **5**A in a first bent configuration.

[0025] FIG. **6**B is a sectional view taken along line **6**B-**6**B of FIG. **6**A.

[0026] FIG. **7**A shows the stapling member of FIG. **5**A in a second bent configuration.

[0027] FIG. **7**B is a sectional view taken along line **7**B-**7**B of FIG. **7**A.

[0028] FIG. **8**A shows a portion of an exemplary stapling member, comprising legs having arrow-shaped tips.

[0029] FIG. **8**B is a sectional view of a stapling portion, taken along line **8**B-**8**B of FIG. **8**A.

[0030] FIG. **8**C shows the stapling portion of FIG. **8**B in a second bent configuration.

[0031] FIG. **9**A is a sectional view of a stapling portion comprising a stapling element, shown in a first bent configuration.

[0032] FIG. **9**B shows the stapling portion of FIG. **9**A in a second bent configuration.

[0033] FIG. **10** shows a leaflet positioned between a strut and a stapling member, prior to being coupled by the stapling member to the strut.

[0034] FIGS. **11**A-**11**B show a leaflet sandwiched between the strut and the stapling member in a first bent configuration of the stapling member.

[0035] FIG. **12** shows the leaflet coupled to the strut by the stapling member shown in FIGS. **11**A-**11**B, in the second bent configuration of the stapling member.

[0036] FIG. **13**A shows an exemplary stapling member in a first bent configuration, wherein the legs of each stapling portion are offset from each other.

[0037] FIG. **13**B shows an enlarged view of one of the stapling portions of FIG. **13**A.

[0038] FIG. **14**A shows the stapling member of FIG. **13**A in a second bent configuration.

[0039] FIG. **14**B shows an enlarged view of one of the stapling portions of FIG. **14**A.
[0040] FIGS. **15**A-**15**B show a leaflet sandwiched between the strut and a stapling member with offset legs, in a first bent configuration of the stapling member.
[0041] FIG. **16** shows the leaflet coupled to the strut by the stapling member shown in FIGS. **15**A-**15**B, in the second bent configuration of the stapling member.
[0042] FIG. **17** is a perspective view of an exemplary prosthetic valve that includes a leaflet structure coupled to the frame by stapling members.
[0043] FIG. **18** shows an exemplary leaflet comprising a stiff portion.
[0044] FIG. **19** shows the leaflet of FIG. **18** coupled to a strut by a stapling member.
[0045] FIG. **20**A shows an exemplary stapling member formed of a continuous wire, with tips of the legs comprised of lateral bent extensions.
[0046] FIG. **20**B shows the stapling member of FIG. **20**A in a first bent configuration.
[0047] FIG. **20**C shows the stapling member of FIG. **20**A in a second bent configuration.
[0048] FIG. **21**A shows an exemplary stapling member formed of a continuous wire, with pointed tips of the legs.
[0049] FIG. **21**B shows the stapling member of FIG. **21**A in a first bent configuration.
[0050] FIG. **21**C shows the stapling member of FIG. **21**A in a second bent configuration.
[0051] FIG. **22** shows a portion of an exemplary frame comprising recessed struts and unrecessed struts.
[0052] FIG. **23** shows a portion of an exemplary frame comprising recessed struts which are wider than unrecessed struts.
[0053] FIG. **24** shows an exemplary compressible insert.
[0054] FIG. **25**A shows a cross-sectional view of a leaflet disposed between a recessed strut and a compressible insert, prior to the leaflet being couple to the strut.
[0055] FIG. **25**B shows a cross-sectional view of a leaflet coupled to the recessed strut by the compressible insert.
[0056] FIG. **26** shows an exemplary surgically implantable prosthetic valve.
[0057] FIG. **27** is an exploded view of components of an exemplary prosthetic valve comprising gripping tubes.
[0058] FIG. **28** shows a leaflet coupled to a frame of an exemplary prosthetic valve by a gripping tube.
[0059] FIG. **29**A shows a cross-sectional view of leaflet coupled to a strut by a gripping tube, with the cusp edge retained within the inner channel of the gripping tube.
[0060] FIG. **29**B shows a cross-sectional view of leaflet coupled to a strut by a gripping tube, with the cusp edge disposed out of the inner channel of the gripping tube.
[0061] FIG. **30** shows an exemplary gripping tube that includes gripping tube cusp portions and gripping tube commissure portions.
[0062] FIG. **31** shows a portion of an exemplary stapling member comprising blunt tips positioned next to an exemplary leaflet comprising preformed perforations.
[0063] FIG. **32**A shows a portion of an exemplary stapling member comprising a single leg in its stapling portion, prior to coupling a leaflet to a strut.
[0064] FIG. **32**B shows the leaflet coupled to the strut by the stapling member of FIG. **32**A, in the second bent configuration of the stapling member.
[0065] FIG. **33** shows a portion of an exemplary stapling member having a zig-zagged spine.

DETAILED DESCRIPTION

[0066] For purposes of this description, certain aspects, advantages, and novel features of the examples of this disclosure are described herein. The disclosed methods, apparatus, and systems should not be construed as being limiting in any way. Instead, the present disclosure is directed toward all novel and nonobvious features and aspects of the various disclosed examples, alone and in various combinations and sub-combinations with one another. The methods, apparatus, and systems are not limited to any specific aspect or feature or combination thereof, nor do the disclosed examples require that any one or more specific advantages be present, or problems be solved. The technologies from any example can be combined with the technologies described in any one or more of the other examples. In view of the many possible examples to which the principles of the disclosed technology may be applied, it should be recognized that the illustrated examples are only preferred examples and should not be taken as limiting the scope of the disclosed technology.

[0067] Although the operations of some of the disclosed examples are described in a particular, sequential order for convenient presentation, it should be understood that this manner of description encompasses rearrangement, unless a particular ordering is required by specific language set forth below. For example, operations described sequentially may in some cases be rearranged or performed concurrently. Moreover, for the sake of simplicity, the attached figures may not show the various ways in which the disclosed methods can be used in conjunction with other methods. Additionally, the description sometimes uses terms like “provide” or “achieve” to describe the disclosed methods. These terms are high-level abstractions of the actual operations that are performed. The actual operations that correspond to these terms may vary depending on the particular implementation and are readily discernible by one of ordinary skill in the art.

[0068] All features described herein are independent of one another and, except where structurally impossible, can be used in combination with any other feature described herein.

[0069] As used in this application and in the claims, the singular forms “a,” “an,” and “the” include the plural forms unless the context clearly dictates otherwise. Additionally, the terms “have” or “includes” means “comprises”. Further, the terms “coupled”, “connected”, and “attached”, as used herein, are interchangeable and generally mean physically, mechanically, chemically, magnetically, and/or electrically coupled or linked and does not exclude the presence of intermediate elements between the coupled or associated items absent specific contrary language. As used herein, “and/or” means “and” or “or”, as well as “and” and “or”.

[0070] Directions and other relative references may be used to facilitate discussion of the drawings and principles herein, but are not intended to be limiting. For example, certain terms may be used such as “inner,” “outer,” “upper,” “lower,” “inside,” “outside,”, “top,” “bottom,” “interior,” “exterior,” “left,” right,” and the like. Such terms are used, where applicable, to provide some clarity of description when dealing with relative relationships, particularly with respect to the illustrated examples. Such terms are not, however, intended to imply absolute relationships, positions, and/or orientations. For example, with respect to an object, an “upper” part can become a “lower” part simply by turning the object over. Nevertheless, it is still the same part and the object remains the same.

[0071] The term “plurality” or “plural” when used together with an element means two or more of the elements. Directions and other relative references (e.g., inner and outer, upper and lower, above and below, left and right, and proximal and distal) may be used to facilitate discussion of the drawings and principles herein but are not intended to be limiting.

[0072] The terms “proximal” and “distal” are defined relative to the use position of a delivery apparatus. In general, the end of the delivery apparatus closest to the user of the apparatus is the proximal end, and the end of the delivery apparatus farthest from the user (e.g., the end that is inserted into a patient's body) is the distal end. The term “proximal” when used with two spatially separated positions or parts of an object can be understood to mean closer to or oriented towards the proximal end of the delivery apparatus. The term “distal” when used with two spatially separated positions or parts of an object can be understood to mean closer to or oriented towards the distal end of the delivery apparatus. The terms “longitudinal” and “axial” are interchangeable, and refer to an axis extending in the proximal and distal directions, unless otherwise expressly defined.

[0073] The terms “axial direction,” “radial direction,” and “circumferential direction” have been used herein to describe the arrangement and assembly of components relative to the geometry of the frame of the prosthetic valve, or the geometry of an inflatable balloon that can be used to expand a prosthetic valve. Such terms have been used for convenient description, but the disclosed examples are not strictly limited to the description. In particular, where a component or action is described relative to a particular direction, directions parallel to the specified direction as well as minor deviations therefrom are included. Thus, a description of a component extending along an axial direction of the frame does not require the component to be aligned with a center of the frame; rather, the component can extend substantially along a direction parallel to a central axis of the frame.

[0074] As used herein, the terms “integrally formed” and “unitary construction” refer to a construction that does not include any welds, fasteners, or other means for securing separately formed pieces of material to each other.

[0075] As used herein, operations that occur “simultaneously” or “concurrently” occur generally at the same time as one another, although delays in the occurrence of operation relative to the other due to, for example, spacing between components, are expressly within the scope of the above terms, absent specific contrary language.

[0076] As used herein, terms such as “first,” “second,” and the like are intended to serve as respective labels of distinct components, steps, etc. and are not intended to connote or imply a specific sequence or priority. For example, unless otherwise stated, a step of performing a second action and/or of forming a second component may be performed prior to a step of performing a first action and/or of forming a first component.

[0077] As used herein, the term “substantially” means the listed value and/or property and any value and/or property that is at least 75% of the listed value and/or property. Equivalently, the term “substantially” means the listed value and/or property and any value and/or property that differs from the listed value and/or property by at most 25%. For example, “at least substantially parallel” refers to directions that are fully parallel, and to directions that diverge by up to 22.5 degrees.

[0078] In the present disclosure, a reference numeral that includes an alphabetic label (for example, “a,” “b,” “c,” etc.) is to be understood as labeling a particular example of the structure or component corresponding to the reference numeral. Accordingly, it is to be understood that components sharing like names and/or like reference numerals (for example, with different alphabetic labels or without alphabetic labels) may share any properties and/or characteristics as disclosed herein even when certain such components are not specifically described and/or addressed herein.

[0079] Throughout the figures of the drawings, different superscripts for the same reference numerals are used to denote different examples of the same elements. Examples of the disclosed devices and systems may include any combination of different examples of the same elements. Specifically, any reference to an element without a superscript may refer to any alternative example of the same element denoted with a superscript. In order to avoid undue clutter from having too many reference numbers and lead lines on a particular drawing, some components will be introduced via one or more drawings and not explicitly identified in every subsequent drawing that contains that component.

[0080] FIGS. **1**A and **1**B show perspective views of one example of a prosthetic valve **100**, with and without an outer skirt **107** surrounding the frame **110**, respectively. FIG. **1**C shows the frame **110** without any other soft components attached thereto. The term “prosthetic valve”, as used herein, refers to any type of a prosthetic valve deliverable to a patient's target site over a catheter, which is radially expandable and compressible between a radially compressed, or crimped, state, and a radially expanded state. Thus, the prosthetic valve can be crimped on or retained by an implant delivery apparatus **200** (shown in FIG. **4**) in the radially compressed state during delivery, and then expanded to the radially expanded state once the prosthetic valve reaches the implantation site. The expanded state may include a range of diameters to which the valve may expand, between the compressed state and a maximal diameter reached at a fully expanded state. Thus, a plurality of partially expanded states may relate to any expansion diameter between radially compressed or crimped state, and maximally expanded state. A prosthetic valve of the current disclosure (e.g., prosthetic valve **100**) may include any prosthetic valve configured to be mounted within the native aortic valve, the native mitral valve, the native pulmonary valve, and the native tricuspid valve.

[0081] It is understood that the prosthetic valves disclosed herein may be used with a variety of implant delivery apparatuses. Balloon expandable valves generally involve a procedure of inflating a balloon within a prosthetic valve, thereby expanding the prosthetic valve within the desired implantation site. Once the valve is sufficiently expanded, the balloon is deflated and retrieved along with a delivery apparatus (not shown). Self-expandable valves include a frame that is shape-set to automatically expand as soon an outer retaining shaft or capsule (not shown) is withdrawn proximally relative to the prosthetic valve. Mechanically expandable valves are a category of prosthetic valves that rely on a mechanical actuation mechanism for expansion. The mechanical actuation mechanism usually includes a plurality of expansion and locking assemblies (such as the prosthetic valves described in U.S. Pat. No. 10,603,165, International Application No. PCT/US2021/052745 and U.S. Provisional Application Nos. 63/085,947 and 63/209,904, each of which is incorporated herein by reference in its entirety), releasably coupled to respective actuation assemblies of a delivery apparatus, controlled via a handle (not shown) for actuating the expansion and locking assemblies to expand the prosthetic valve to a desired diameter. The expansion and locking assemblies may optionally lock the valve's diameter to prevent undesired recompression thereof, and disconnection of the actuation assemblies from the expansion and locking assemblies, to enable retrieval of the delivery apparatus once the prosthetic valve is properly positioned at the desired site of implantation.

[0082] FIGS. **1**A-**1**C show an example of a prosthetic valve **100**, which can be a balloon expandable valve, illustrated in an expanded state. The prosthetic valve **100** can comprise an outflow end **101** and an inflow end **102**. In some instances, the outflow end **101** is the proximal end of the prosthetic valve **100**, and the inflow end **102** is the distal end of the prosthetic valve **100**. Alternatively, depending for example on the delivery approach of the valve, the outflow end can be the distal end of the prosthetic valve, and the inflow end can be the distal end of the proximal valve.

[0083] The term “outflow”, as used herein, refers to a region of the prosthetic valve through which the blood flows through and out of the prosthetic valve **100**.

[0084] The term “inflow”, as used herein, refers to a region of the prosthetic valve through which the blood flows into the prosthetic valve **100**.

[0085] In the context of the present application, the terms “lower” and “upper” are used interchangeably with the terms “inflow” and “outflow”, respectively. Thus, for example, the lower end of the prosthetic valve is its inflow end and the upper end of the prosthetic valve is its outflow end.

[0086] In the context of the present application, the terms “lower” and “upper” are used interchangeably with the terms “distal to” and “proximal to”, respectively. Thus, for example, a lowermost component can refer to a distal-most component, and an uppermost component can similarly refer to a proximal-most component.

[0087] The prosthetic valve **100** comprises an annular frame **110** movable between a radially compressed configuration and a radially expanded configuration, and a valvular structure **160** mounted within the frame **110**. The frame **110** can be made of various suitable materials, including plastically-deformable materials such as, but not limited to, stainless steel, a nickel based alloy (e.g., a cobalt-chromium or a nickel-cobalt-chromium alloy such as MP35N alloy), polymers, or combinations thereof. When constructed of a plastically-deformable materials, the frame **110** can be crimped to a radially compressed state on a balloon catheter, and then expanded inside a patient by an inflatable balloon or equivalent expansion mechanism. Alternatively or additionally, the frame **110** can be made of shape-memory materials such as, but not limited to, nickel titanium alloy (e.g., Nitinol). When constructed of a shape-memory material, the frame **110** can be crimped to a radially compressed state and restrained in the compressed state by insertion into a shaft or equivalent mechanism of a delivery apparatus.

[0088] In the example illustrated in FIGS. **1**A-**1**C, the frame **110** is an annular, stent-like structure comprising a plurality of intersecting struts **114**. In this application, the term “strut” encompasses axial struts, angled struts, laterally extendable struts, commissure windows, commissure support struts, support posts, and any similar structures described by U.S. Pat. Nos. 7,993,394 and 9,393,110, which are incorporated herein by reference. A strut **114** may be any elongated member or portion of the frame **110**. The frame **110** can include a plurality of strut rungs that can collectively define one or more rows of cells **130**. The frame **110** can have a cylindrical or substantially cylindrical shape having a constant diameter from the inflow end **102** to the outflow end **101** as shown, or the frame can vary in diameter along the height of the frame, as disclosed in U.S. Pat. No. 9,155,619, which is incorporated herein by reference.

[0089] The end portions of the struts **114** are forming apices **128** at the outflow end **101** and apices **129** at the inflow end **102**. The struts **114** can intersect at additional junctions **127** formed between the outflow apices **128** and the inflow apices **129**. The junctions **127** can be equally or unequally spaced apart from each other, and/or from the apices **128**, **129**, between the outflow end **101** and the inflow end **102**.

[0090] The struts **114** can include a plurality of angled struts **115** and vertical or axial struts **116**. FIGS. **1**A-**1**C show an exemplary prosthetic valve **100** that can be representative of, but is not limited to, a balloon expandable prosthetic valve. The frame **110** of the prosthetic valve **100** illustrated in FIG. **1**C comprises rungs of angled struts **115** and axial struts **116** disposed between some of the rungs of the angled struts. In such implementations of the frame, the struts can be pivotable or bendable relative to each other, so as to permit frame expansion or compression. For example, the frame **110** can be formed from a single piece of material, such as a metal tube, via various processes such as, but not limited to, laser cutting, electroforming, and/or physical vapor deposition, while retaining the ability to collapse/expand radially in the absence of hinges and like.

[0091] Each strut **114** can define a strut inner surface **112** facing the valve central longitudinal axis Ca, and an opposite outer surface **113** facing away from the axis Ca. The inner surfaces **112** of the struts **114** collectively define an inner surface of the frame **110**. The outer surfaces **113** of the struts **114** collectively define an outer surface of the frame **110**. In some implementations, a strut **114** can be formed to have a rectangular cross-section, defining a struts width W.sub.S and a strut thickness Ts (shown, for example, in FIG. **10**). Strut thickness Ts is measured between the strut inner surface **112** and the strut outer surface **113**. Strut width Ws is measured between the opposing surfaces of the strut **114** that extend between its inner **112** and outer **113** surfaces.

[0092] A conventional valvular structure **160**, shown also for example in FIG. **2**, can include a plurality of leaflets **162** (e.g., three leaflets), positioned at least partially within the frame **110**, and configured to regulate flow of blood through the prosthetic valve **100** from the inflow end **102** to the outflow end **101**. While three leaflets **162** arranged to collapse in a tricuspid arrangement, are shown in the example illustrated in FIGS. **1**A-**1**B and **2**, it will be clear that a prosthetic valve **100** can include any other number of leaflets **162**. Adjacent leaflets **162** can be arranged together to form commissures **180** that are coupled (directly or indirectly) to respective portions of the frame **110**, thereby securing at least a portion of the valvular structure **160** to the frame **110**. In some examples, at least some (e.g., three) of the axial struts **116** can define axially extending window frame portions, also termed commissure windows **119**, configured to mount respective commissures **180** of the valvular structure **160**. The leaflets **162** can be made from, in whole or part, biological material (e.g., pericardium), bio-compatible synthetic materials, or other such materials. Further details regarding transcatheter prosthetic heart valves, including the manner in which the valvular structures **160** can be coupled to the frame **110** of the prosthetic valve **100**, can

be found, for example, in U.S. Pat. Nos. 6,730,118, 7,393,360, 7,510,575, 7,993,394, 8,652,202, and 11,135,056, all of which are incorporated herein by reference in their entireties.
[0093] As shown for example in FIG. **2**, three separate leaflets **162** can collectively define the valvular structure **160** in some cases. Each conventional separate leaflet **162** can have a rounded cusp edge **164** opposite a free edge **166**, and a pair of generally oppositely-directed tabs **168** separating the cusp edge **164** and the free edge **166**. The cusp edge **164** in such cases forms a single scallop. A cusp end portion **165** can be defined for each leaflet **162** as the portion extending from the cusp edge **164**, and can be rounded so as to generally track the shape of the cusp edge **164**. Each separate leaflet **162** further comprises an inner surface (not annotated), defined as the surface facing the valve central longitudinal axis Ca, and an outer surface (not annotated), opposite thereto so as to face the frame **110**. The leaflet **162** can define a leaflet thickness T.sub.L, measured between the leaflet's inner and outer surfaces.
[0094] When such leaflets **162** are coupled to the frame and to each other, the lower edge of the resulting valvular structure **160** desirably has an undulating, curved scalloped shape. By forming the leaflets with this scalloped geometry, stresses on the leaflets **162** are reduced which, in turn, improves durability of the prosthetic valve. Moreover, by virtue of the scalloped shape, folds and ripples at the belly of each leaflet, which can cause early calcification in those areas, can be eliminated or at least minimized. The scalloped geometry also reduces the amount of tissue material used to form the valvular structure, thereby allowing a smaller, more even crimped profile at the inflow end of the valve.
[0095] The leaflets **162** define a non-planar coaptation plane (not annotated) when their free edges **166** co-apt with each other to seal blood flow through the prosthetic valve **100**. Leaflets **162** can be secured to one another at their tabs **168** to form commissures **180** of the valvular structure **160**, which can be secured, directly or indirectly, to structural elements connected to the frame **110** or integrally formed as portions thereof, such as commissure posts, commissure windows, and the like. When secured to two other leaflets **162** to form valvular structure **160**, the cusp edges **164** of the leaflets **162** collectively form the scalloped line **105** of the valvular structure **160**. Each leaflet **162** comprises a leaflet body **170**, defined between the line of attachment of the leaflet to the frame, for example along scalloped line **105**, and the free edge **166**. The leaflet body **170** defines the movable portion of the leaflet **162**, free to move toward the frame **110** in an open state of the valvular structure **160**, and toward central longitudinal axis Ca to co-apt with other leaflets **162** in a closed state of the valvular structure **160**. The cusp end portions **165** of leaflets **162** can be defined as the portions along which the leaflet **162** are coupled to the frame **110** along the cusp edges **164**, collectively forming scalloped line **105** of the valvular structure **160**, such that the leaflet body **170** can be defines as the movable portion of the leaflet **162** extending from the cusp end portions **165** to the free edge **166**.
[0096] Various exemplary implementations for various components and devices can be referred to, throughout the specification, with superscripts, for ease of explanation of features that refer to such exemplary implementations. It is to be understood, however, that any reference to structural or functional features of any apparatus, device or component, without a superscript, refer to these features being commonly shared by all specific exemplary implementations that can be also indicated by superscripts. In contrast, features emphasized with respect to an exemplary implementation of any assembly, apparatus or component, referred to with a superscript, may be optionally shared by some but not necessarily all other exemplary implementations. For example, prosthetic valve **100**.sup.a is an exemplary implementation of prosthetic valve **100**, and thus includes all of the features described for prosthetic valve **100** throughout the current disclosure, except that while the leaflets **162** of a prosthetic valve **100** can be attached to the frame **110** along their cusp end portions **165** in any manner, the leaflets **162** of prosthetic valve **100**.sup.a are sutured to an inner skirt, which is in turn attached to the frame of the valve **100**.sup.a.
[0097] In some examples, the prosthetic valve **100** can further comprise at least one skirt or sealing

member. FIGS. **1**A-**1**B show an example of a prosthetic valve **100**.sup.a that includes an inner skirt **106**, which can be secured to the inner surface of the frame **110**. Such an inner skirt **106** can be configured to function, for example, as a sealing member to prevent or decrease perivalvular leakage. An inner skirt **106** can further function as an anchoring region for valvular structure **160**.sup.a to the frame **110**, and/or function to protect the leaflets **162** against damage which may be caused by contact with the frame **110**, for example during valve crimping or during working cycles of the prosthetic valve **100**. FIG. **1**B shows an inner skirt **106** disposed around and attached to the inner surface of frame **110**, wherein the valvular structure **160**.sup.a is sutured to the inner skirt **106** along scalloped line **105**. Additionally, or alternatively, the prosthetic valve **100** can comprise an outer skirt **107** mounted on the outer surface of frame **110**, configure to function, for example, as a sealing member retained between the frame **110** and the surrounding tissue of the native annulus against which the prosthetic valve is mounted, thereby reducing risk of paravalvular leakage (PVL) past the prosthetic valve **100**.
[0098] Any of the inner skirt **106** and/or outer skirt **107** can be made of various suitable biocompatible materials, such as, but not limited to, various synthetic materials (e.g., PET) or natural tissue (e.g., pericardial tissue). In some cases, the inner skirt **106** can be formed of a single sheet of material that extends continuously around the inner surface of frame **110**. In some cases, the outer skirt **107** can be formed of a single sheet of material that extends continuously around the outer surface of frame **110**.
[0099] FIG. **3** shows an exemplary prosthetic valve **100**.sup.b. Prosthetic valve **100**.sup.b is an exemplary implementation of prosthetic valve **100** and/or **100**.sup.a, and thus includes all of the features described for prosthetic valve **100** and/or **100**.sup.a throughout the current disclosure, except that instead of the valvular structure **160**.sup.a of prosthetic valve **100**.sup.a anchored to an inner skirt **106**, which in turn is coupled to the frame **110**, the valvular structure **160**.sup.b of prosthetic valve **100**.sup.b is sutured directly to the frame **110**.sup.b. For example, as shown in FIG. **3**, each leaflet **162** can be directly sutured to struts **114** of the frame **110**.sup.b by one or more sutures **184**, instead of via inner skirt **106**. While not shown in FIG. **3**, the prosthetic valve **100**.sup.b can further include an outer skirt **107** disposed around the frame **110**, similarly to prosthetic valve **100**.sup.a.
[0100] The frame **110**.sup.b shown in FIG. **3** can comprise a plurality of struts **114**, such as angled struts **115**, including a group of first struts **115***a* to which the leaflets **162** are not sutured along their cusp end portions **165**, and a group of second struts **115***b* to which the cusp end portions **165** of the leaflets **162** are sutured, forming a scalloped line **105** along selected angled struts **115***b*.
[0101] FIG. **4** illustrates a delivery apparatus **200**, according to an exemplary configuration, adapted to deliver a prosthetic valve **100** according to any example described herein. It should be understood that the delivery apparatus **200** can be used to implant prosthetic devices other than prosthetic valves, such as stents or grafts.
[0102] The delivery apparatus **200** includes a handle **204** and a balloon catheter **252** having an inflatable balloon **250** mounted on its distal end. The prosthetic valve **100** can be carried in a crimped state over the balloon catheter **252**. Optionally, an outer delivery shaft **224** can concentrically extend over the balloon catheter **252**, and a push shaft **220** can be disposed over the balloon catheter **252**, optionally between the balloon catheter **252** and the outer delivery shaft **224**.
[0103] The outer delivery shaft **224**, the push shaft **220**, and the balloon catheter **252**, can be configured to be axially movable relative to each other. For example, a proximally oriented movement of the outer delivery shaft **224** relative to the balloon catheter **252**, or a distally oriented movement of the balloon catheter **252** relative to the outer delivery shaft **224**, can expose the prosthetic valve **100** from the outer delivery shaft **224**. The delivery apparatus **200** can further include a nosecone **240** carried by a nosecone shaft (hidden from view in FIG. **4**) extending through a lumen of the balloon catheter **252**.
[0104] The proximal ends of the balloon catheter **252**, the outer delivery shaft **224**, the push shaft

220, and optionally the nosecone shaft, can be coupled to the handle **204**. During delivery of the prosthetic valve **100**, the handle **204** can be maneuvered by an operator (e.g., a clinician or a surgeon) to axially advance or retract components of the delivery apparatus **200**, such as the nosecone shaft, the balloon catheter **252**, the outer delivery shaft **224**, and/or the push shaft **220**, through the patient's vasculature, as well as to inflate the balloon **250** mounted on the balloon catheter **252**, so as to expand the prosthetic valve **100**, and to deflate the balloon **250** and retract the delivery apparatus **200** once the prosthetic valve **100** is mounted in the implantation site.

[0105] The handle **204** can include a steering mechanism configured to adjust the curvature of the distal end portion of the delivery apparatus **200**. In the illustrated example, the handle **204** includes an adjustment member, such as the illustrated rotatable knob **206***a*, which in turn is operatively coupled to the proximal end portion of a pull wire. The pull wire can extend distally from the handle **204** through the outer delivery shaft **224** and has a distal end portion affixed to the outer delivery shaft **224** at or near the distal end of the outer delivery shaft **224**. Rotating the knob **206***a* can increase or decrease the tension in the pull wire, thereby adjusting the curvature of the distal end portion of the delivery apparatus **200**. Further details on steering or flex mechanisms for the delivery apparatus can be found in U.S. Pat. No. 9,339,384, which is incorporated by reference herein. The handle **204** can further include an adjustment mechanism including an adjustment member, such as the illustrated rotatable knob **206***b*. The adjustment mechanism can be configured to adjust the axial position of the push shaft **220** relative to the balloon catheter.

[0106] The prosthetic valve **100** can be carried by the delivery apparatus **200** during delivery in a crimped state, and expanded by balloon inflation to secure it in a native heart valve annulus. In an exemplary implantation procedure, the prosthetic valve **100** is initially crimped over the balloon catheter **252**, proximal to the inflatable balloon **250**. Because prosthetic valve **100** is crimped at a location different from the location of balloon **250**, prosthetic valve **100** can be crimped to a lower profile than would be possible if it was crimped on top of balloon **250**. This lower profile permits the clinician to more easily navigate the delivery apparatus **200** (including crimped prosthetic valve **100**) through a patient's vasculature to the treatment location. The lower profile of the crimped prosthetic valve is particularly helpful when navigating through portions of the patient's vasculature which are particularly narrow, such as the iliac artery.

[0107] The balloon **250** can be secured to balloon catheter **252** at its balloon proximal end, and to either the balloon catheter **252** or the nosecone **240** at its distal end. The distal end portion of the push shaft **220** is positioned proximal to the outflow end **101** of the prosthetic valve **100**.

[0108] When reaching the site of implantation, and prior to balloon inflation, the push shaft **220** is advanced distally, allowing its distal end portion to contact and push against the outflow end of prosthetic valve **100**, pushing the valve **100** distally therewith. The distal end of push shaft **220** is dimensioned to engage with the outflow end of the prosthetic valve **100** in a crimped configuration of the valve. In some implementations, the distal end portion of the push shaft **220** can be flared radially outward, to terminate at a wider-diameter that can contact the prosthetic valve **100** in its crimped state. Push shaft **220** can then be advanced distally, pushing the prosthetic valve **100** therewith, until the crimped prosthetic valve **100** is disposed around the balloon **250**, at which point the balloon **250** can be inflated to radially expand the prosthetic valve **100**. Once the prosthetic valve **100** is expanded to its functional diameter within a native annulus, the balloon **250** can be deflated, and the delivery apparatus **200** can be retrieved from the patient's body.

[0109] In particular implementations, the delivery apparatus **200** with the prosthetic valve **100** assembled thereon, can be packaged in a sterile package that can be supplied to end users for storage and eventual use. In particular implementations, the leaflets of the prosthetic valve (typically made from bovine pericardium tissue or other natural or synthetic tissues) are treated during the manufacturing process so that they are completely or substantially dehydrated and can be stored in a partially or fully crimped state without a hydrating fluid. In this manner, the package containing the prosthetic valve **100** and the delivery apparatus **200**, can be free of any liquid.

Methods for treating tissue leaflets for dry storage are disclosed in U.S. Pat. Nos. 8,007,992 and 8,357,387, both of which documents are incorporated herein by reference.
[0110] Conventional prosthetic valves, such as valve **100**.sup.a or **100**.sup.b, include a valvular structure **160** with the leaflets **162** that can be either sutured to an inner skirt **106**, as illustrated and described for a prosthetic valve **100**.sup.a, or sutured directly to struts **114** of frame **110**, as illustrated and described for a prosthetic valve **100**.sup.b. The suturing process tends to be labor-intensive, and the quality of the result may depend on the skill-level of the assembler. Moreover, the tension applied to the suture may not be well controlled, which can effect local geometries of the leaflets **162**. All of these factors may adversely impact the functionality and structural integrity of the valvular structure **160** in the long term. Disclosed herein are devices and method for securing the leaflets **162**, such as along their cusp end portions **165**, to the frame **110** of prosthetic valve **100**, in a manner that obviates the use of sutures stitched through the leaflet material, which can substantially reduce process times and the difficulty valve assembly procedures. In some examples, the term “cusp end portion” refers to a portion of the leaflet extending to a distance of about twice the strut width W.sub.S. In some examples, the term “cusp end portion” refers to a portion of the leaflet extending to a distance of about three times the strut width W.sub.S. In some examples, the term “cusp end portion” refers to a portion of the leaflet extending to a distance of about four times the strut width W.sub.S.
[0111] FIG. **5**A shows a portion of an exemplary stapling member **300** that can be used for coupling a leaflet **162** to one or more struts **114** of a frame **110**, illustrated in a flattened configuration. FIG. **5**B is a sectional view taken along line **5**B-**5**B of FIG. **5**A. Stapling member **300** comprises a spine **302** extending along stapling member axis Cs, and a plurality of stapling portions **310** separated from each other by intermediate portions **316** of the spine **302**. The spine **302** defines a spine first edge **308***a*, a spine second edge **308***b*, a spine inner surface **304** extending between the spine first and second edges **308***a*, **308***b*, and a spine outer surface **306** opposite to the spine inner surface **304**.
[0112] Each stapling portion **310** comprises two legs **318** extending from the opposite edges **308** of the spine, such as a first leg **318***a* extending from the spine first edge **308***a* and a second leg **318**.sup.b extending from the spine second edge **308***b*, and a backspan **314** extending between the first and second legs **318***a*, **318**.sup.b. The backspan can be perpendicular to the stapling member axis Ca, such that both legs **318***a*, **318**.sup.b extend from the backspan at opposite sides of the stapling member axis Ca. FIG. **5**C shows an enlarged view of an end portion of a leg **318**. Each leg **318** can have a wide W.sub.L defined in a direction which is generally parallel to the direction of stapling member axis Ca. Each leg **318** terminates at a tip **330**, which can be a relatively sharp tip configured to penetrate through the thickness of a leaflet **162**, such as through a pericardial tissue the leaflet **162** can be formed of. As shown in FIG. **5**B, the backspan **314** can define a backspan width W.sub.B between both of the spine edges **308***a*, **308***b*. Each leg **318** can have a length L.sub.L between the corresponding spine edge **308** and the tip **330**. In some examples, the backspan width W.sub.B is at least as great as the strut width W.sub.S.
[0113] In some examples, the tip **330** can define a slanted or beveled surface. FIGS. **5**A-**5**C show an exemplary stapling member **300**.sup.a, which is an exemplary implementation of stapling member **300**, and thus includes all of the features described for stapling member **300** throughout the current disclosure, except that the tip **330**.sup.a defines a slanted surface between the spine inner **304** and outer **306** surface.
[0114] Stapling member **300** can be formed, in some examples, as a unitary component, such as be being stamped, laser cut, and the like, from a plate, to form the desired shape of the stapling member **300**. In such examples, the spine **302** (including intermediate portions **316** thereof) and the stapling portions **310** are integrally formed. FIG. **5**A shows an optional form the stapling member **300** can be provided in after being cut (e.g., laser cut), wherein the legs **318** are co-planar along their entire lengths L.sub.L with the spine **302** and/or the intermediate portions **316**. While the legs

318 and the spine **302** are shown to have rectangular cross-sectional shapes, it is to be understood that any of the legs **318**, the spine **302**, the intermediate portions **316**, and/or backspans **314**, can have any other cross-sectional shape, such as square shape, rhomboid, triangular, circular, elliptical, and the like. In some implementations, cross-sectional shapes of the spine **302** and/or backspans **314** defining a flat spine outer surface **306** can be advantageous in that the flat surface provides increased contact area with the leaflet **162**, when pressed against the leaflet to couple it to the frame, as opposed to round surface configurations.

[0115] FIG. **6**A shows the stapling member **300** in a first bent configuration, wherein the legs **318** are bent at first bends **322** relative to backspans **314**. FIG. **6**B is a sectional view taken along line **6**B-**6**B of FIG. **6**A. As shown, each leg **318** is bent at a first bend **322**, relative to the backspan **314**, for example at a right angle. The legs **318** are bend outwards, away from the spine inner surface **304**, such that each stapling portion **310** can be generally U-shaped. While shown to bend at right angles, such that both legs **318***a*, **318**.sup.b are generally parallel to each other between first bends **322** and tips **330**, it is to be understood that in some implementations, the legs **318** can be bent at other angles, which can be more or less than 90°.

[0116] It is to be understood that any reference throughout the current disclosure to right angles, may refer not only to 90°, but rather to angles approximating this value within a range of ±15°, meaning that any reference to a right angle can refer to an angle in the range of 75°-105°.

[0117] Thus, in some implementations, the stapling member **300** can be manufactured from flat metal stock, as shown in FIG. **5**A, and then bent to the first bent configuration, as shown in FIG. **6**A. The stapling member **300** can be made of plastically-deformable materials such as, but not limited to, stainless steel or a nickel based alloy (e.g., a cobalt-chromium or a nickel-cobalt-chromium alloy such as MP35N alloy), titanium, and the like.

[0118] FIG. **7**A shows the stapling member **300** in a second bent configuration, wherein the legs **318** are further bent towards each other at second bends **324**. FIG. **7**B is a sectional view taken along line **7**B-**7**B of FIG. **6**A. As shown, each leg is bent at a second bend **324**, between the first bend **322** and the tip **330**, dividing the leg **318** into a radial portion **326** extending between the first bend **322** and the second bend **324**, and a clinch portion **328** extending between the second bend **324** and the tip **330**. The radial portion **326** can extend along a radial portion length L.sub.R, and the clinch portion **328** can extend along a clinch portion length L.sub.C, such that L.sub.R+L.sub.C=L.sub.L.

[0119] The first and second clinch portions **328***a*, **328**.sup.b extend towards each other, and can be generally parallel to the backspan **314**. In some examples, as shown in FIGS. **5**A-**7**B, the legs **318** of each stapling portion **310** are aligned with each other, such that their tips **330** are positioned in front of each other in the second bent configuration. In such cases, since the clinch portions **328** are coaxial, it may be desirable to design the legs **318** such that the combined sum of lengths L.sub.C of both clinch portions **328** is equal to or less than the backspan width W.sub.B and/or less to prevent them from engaging each other when bent to the second bent configuration. In some examples, the combined sum of lengths L.sub.C of both clinch portions **328** is equal to or less than the strut width W.sub.S.

[0120] In some examples, the tip **330** can define an arrowhead shape. FIG. **8**A shows a portion of an exemplary stapling member **300**.sup.b in a first bent configuration. FIG. **8**B is a sectional view taken along line **8**B-**8**B of FIG. **8**A, showing a stapling portion **310**.sup.b in the first bent configuration. FIG. **8**C shows the stapling portion **310**.sup.b of FIG. **8**B in a second bent configuration. Stapling member **300**.sup.b is an exemplary implementation of stapling member **300**, and thus includes all of the features described for stapling member **300** throughout the current disclosure, except that the tip **330**.sup.b of each leg **318**.sup.b defines an arrowhead shape.

[0121] In some examples, a stapling member **300** can include stapling elements **312** coupled to the spine **302**. FIG. **9**A is a sectional view showing a stapling portion **310**.sup.c of a stapling member **300**.sup.c in a first bent configuration. FIG. **9**B shows the stapling portion **310**.sup.c of FIG. **9**A in a second bent configuration. As shown, stapling portion **310**.sup.e comprises a stapling element **312** that includes the backspan **314**.sup.c and legs **318**.sup.c, wherein the stapling portion **310**.sup.c is attached, at its backspan **314**.sup.c, to the spine **302**.sup.c. The spine **302**.sup.c and stapling elements **312** can be provided as separate components, which can be attached to each other by gluing, welding, and the like. The spine **302**.sup.c and stapling elements **312** can be formed of different or similar materials. Providing separately formed stapling elements **312** and spine **302** attachable to each other allows each to be formed with a different shape and cross-section. For example, while spine **302**.sup.c can be manufactured (e.g., stamped, laser cut, and the like) from a plate (e.g., metal plate), having a rectangular cross section as shown in FIGS. **9**A-**9**B, each stapling element **312** can be formed from a wire (e.g., metal wire) having a circular cross-section. In some examples, the tip **330** can conical, as shown for tips **330**.sup.c in FIGS. **9**A-**9**B.

[0122] FIGS. **10-12** show steps in a method of coupling a leaflet **162** to strut **114** by stapling member **300**. As shown in FIG. **10**, a leaflet's cusp end portion **165** can be placed adjacent to one or more struts **114** of the frame **110** (a portion of one leaflet **162** next to a portion of one strut **114** is shown in FIG. **10**), such that the cusp edge **164** can be somewhat lower than the strut **114**. The leaflet's cusp end portion **165** is placed against the inner surface of the frame **110**, such that the leaflet **162** is facing the strut inner surface **112**. Stapling member **300**, in a first bent configuration thereof, can be then approximated to the leaflet **162** from the inner side, such that the spine inner surface **304** faces the valve central longitudinal axis Ca and the spine outer surface **306** faces the leaflet **162**, orienting the legs **318** toward the leaflet **162**. As shown in FIGS. **11**A-**11**B, illustrated as viewed from the inner and outer sides of the valve frame, the legs **318** are then pressed into the leaflet **162**, causing the tips **330** to pierce through the tissue material, and the legs **318** penetrate into leaflet **162** and extend outward beyond the strut outer surface **113**, sandwiching the leaflet **162** between the backspan **314** and the strut **114**.

[0123] FIG. **12** shows the next step of bending the legs **318** over the outer surface **113** of the strut **114**, transitioning the stapling member **300** to the second bent configuration, thus clamping the stapling member **300** over strut **114** to couple the leaflet **162** to the frame **110**, for example through the cusp end portions **165**. While a stapling member **300**.sup.a is illustrated in FIGS. **10-12**, it is to be understood that this is shown by way of illustration and not limitation, and that a stapling member **300** that includes any other shape of tips **330**, such as stapling members **300**.sup.b or **300**.sup.c described above, can be similarly used.

[0124] In some implementations, as shown for example in FIG. **5**A, the stapling member **300** can be generally curved, meaning that it extends along a curved stapling member axis Cs, optionally following a curvature of the cusp edge **164** and/or cusp end portion **165** of leaflet **162**. In some implementation, the curved shape of one or more stapling member **300** can follow a scalloped shape of the scalloped line **105**, to allow coupling of the valvular structure **160**, by one or more stapling members **300**, along a generally scalloped-shaped line **105**. Intermediate portion **316** can extend along an intermediate portion length L.sub.I between adjacent stapling portions **310**. Intermediate portion length L.sub.I can be measured along the spine first edge **308***a*, the spine second edge **308***b*, or the stapling member axis Cs centrally disposed between the spine edges **308***a* and **308***b*. Different intermediate portions **316** of the spine **302** can have similar or different lengths L.sub.I. For example, intermediate portions **316** extending between subsequent stapling portion **310** configured to couple a leaflet **162** to a section of a strut **114** extending between two neighboring junctions **127** can be shorter in length than intermediate portions **316** that extend along junctions **127** of the frame.

[0125] As shown in FIG. **12**, the radial portions **326** of legs **318** extend through the thickness of the leaflet **162** and across the strut **114**. In some examples, the radial portion length L.sub.R is substantially equal to the combined thicknesses of the leaflet **162** and the strut **114** (i.e., L.sub.R≈T.sub.L+T.sub.S). In some examples, the radial portion length L.sub.R is within the range of ±20% of the combined thicknesses of the leaflet **162** and the strut **114**, such that 0.8

(T.sub.L+T.sub.S)<L.sub.R<1.2 (T.sub.L+T.sub.S).

[0126] As shown, the directions of attachment, extending the legs **318** from the inner side of the frame **110** outwards, may be advantageous, resulting in the relatively sharp tips **330** disposed on the outer side of the valve **100**, which reduces the risk of these tips **330** being otherwise engaged by movable portions of the leaflets **162** inside the valve **100**.

[0127] In some examples, the legs **318** of each stapling portion **310** can be offset from each other. FIG. **13**A shows a portion of a stapling member **300**.sup.d in a first bent configuration. FIG. **13**B shows an enlarged view of one of the stapling portions **310**.sup.d of FIG. **13**A. FIG. **14**A shows the stapling member **300**.sup.d in a second bent configuration. FIG. **14**B shows an enlarged view of one of the stapling portions **310**.sup.d of FIG. **14**A. Stapling member **300**.sup.d is an exemplary implementation of stapling member **300**, and thus includes all of the features described for stapling member **300** throughout the current disclosure, except that the legs **318**.sup.d of each stapling portion **310**.sup.d are not aligned with each other, as shown in FIGS. **5**A-**12**, but are rather offset from each other, such that when the legs **318**.sup.d are bent at second bends **324**, the clinch portion **328**.sup.d do not intersect with each other but rather extend parallel to each other.

[0128] FIGS. **15**A-**16** show steps in a method of coupling a leaflet **162** to strut **114** by stapling member **300**.sup.d. As shown in FIGS. **15**A-**15**B, illustrated as viewed from the inner and outer sides of the valve frame, the tips **330**.sup.d can pierce through the leaflet, such that the legs **318**.sup.d extend outward through the leaflet **162** and beyond the strut outer surface **113**, sandwiching the leaflet **162** between the backspan **314**.sup.d and the strut **114**. As further shown in FIG. **16**, the offset legs **318**.sup.d are then bent over the outer surface **113** of the strut **114**, transitioning the stapling member **300**.sup.a to the second bent configuration, thus clamping the stapling member **300**.sup.d over strut **114** to couple the leaflet **162** to the frame **110**. While a stapling member **300**.sup.d is illustrated in FIGS. **13**A-**16** to include tips **330**.sup.d which are illustrated to include slanted surface, similar to tips **330**.sup.a, it is to be understood that any other shape of tips can be used with offset legs **3184**, including tips forming an arrowhead shape, as illustrated for tip **330**.sup.b, or conical tips **330**.sup.c. Similarly, the offset legs **318**.sup.d can be similarly implemented as offset legs of stapling elements **312** connected to a spine, as described above for stapling members **300**.sup.c.

[0129] Unlike legs **318** which are aligned with each other, limiting the clinch portion length L.sub.C as described above, the offset configuration of legs **318**.sup.d allows the clinch portion **328**.sup.d to be longer, such that the clinch portion length L.sub.C can be, for example, up to the value of the backspan width W.sub.B, without the risk of the tips **330**.sup.d contacting or otherwise interfering with each other. This, in turn, can increase surface contact area between the clinch portions **328**.sup.d and strut **114**, to improve clamping force.

[0130] While described to be equipped with tips **330** shaped to penetrate through the thickness of a leaflet **162**, in some examples, a coupling member **300** can be equipped with blunt tips. FIG. **31** shows a cusp portion of an exemplary portion of a leaflet **1624**, positioned between a strut **114** and exemplary stapling member **300**.sup.e. Stapling member **300**.sup.e is an exemplary implementation of stapling member **300**, and thus includes all of the features described for stapling member **300** throughout the current disclosure, except that the tips **330**.sup.e are blunt. The leaflet **162**.sup.e includes all of the features described for any leaflet **162** throughout the current disclosure, except that the leaflet **162**.sup.d further comprises pre-formed perforations sized to allow passage of the arms **318** therethrough, such that arms **318**.sup.e equipped with blunt tips **330**.sup.e can extend through the leaflet **162**.sup.a without the need to perforate or cut through the leaflet material. It is to be understood that blunt tips, such as shown for tips **330**.sup.e, can be implemented for any other example of stapling member **300** disclosed herein.

[0131] While stapling portions **310** of various exemplary stapling members **300** are described and shown to include two legs **318**, in some examples, a stapling portion **310** can be equipped with a single leg **318**. FIG. **32**A shows an exemplary stapling member **300**.sup.f positioned next to a leaflet **162** prior to coupling the leaflet to the strut **114**. Stapling member **300**.sup.f is an exemplary implementation of stapling member **300**, and thus includes all of the features described for stapling member **300** throughout the current disclosure, except that the stapling portion **310**.sup.f includes a single leg **318**.sup.f instead of two opposing legs. The stapling portion **310**.sup.e is shown in a first bent configuration in FIG. **32**A, with the leg **318**.sup.f bent at a first bend **322**, forming an L-shaped stapling portion **310**.sup.f in this first bent configuration.

[0132] FIG. **32**B shows the stapling member **300**.sup.f of FIG. **32**A coupling the leaflet **162** to the strut **114**, with the leg **318**.sup.f wrapped around the strut **114** in the second bent configuration. In some examples, the leg length L.sub.L can be greater than the sum of the strut thickness T.sub.S and the struts width W.sub.S (i.e., L.sub.L>T.sub.S+W.sub.S), such that the leg **318**.sup.f, in the second bent configuration, can be wrapped around three surfaces of the strut **114**, forming not only a second bend **324** but a third bend **332** as well. As shown in FIG. **32**B, leg **318**.sup.f extends radially inwards, along a radial portion **326** thereof, from the first bend **322** (shown in FIG. **32**A), through the leaflet **162**, toward the strut inner surface **112**. The leg **318**.sup.f is then bent at a second bend **324** to form an axial clinch section **334** extending axially along the strut inner surface **112**, and is finally bent at a third bend **332** to extend radially outward (i.e., toward leaflet **162** or strut outer surface **113**), forming a radial clinch section **336** extending from the third bend **332** and terminating at the tip **330**.

[0133] As shown, the tip **300** is facing the leaflet **162** but does not necessarily contact it or re-penetrate it. This means, in such examples, that the leg length L.sub.L can be less than the sum of strut width W.sub.S and twice the strut thickness T.sub.S (i.e., L.sub.L<2 T.sub.S+W.sub.S), resulting in a radial clinch section **336** which is parallel to, but shorter than, the radial portion **326**. While a second bent configuration forming a third bend **332** that divides the clinch portion **328** into a clinch axial section **334** and a clinch radial section **336** is shown and described with respect to a stapling portion **310**.sup.f having a single leg **318**.sup.f, it is to be understood that other stapling portions **310** described herein to include two legs **310**, such as stapling portion **310**.sup.d of stapling members **300**.sup.d that include pairs of offset legs **3184**, can have leg lengths L.sub.L and be bent in a similar manner to that described above with respect to leg **318**.sup.f. In such cases, the offset nature of a pair of legs allows each to be formed with a length that allows it to be wrapped around three surfaces of the strut, without the legs **318** contacting or otherwise engaging each other. In some examples, a stapling member **300** can be equipped with pairs of legs **318** which are aligned with each other, yet having leg lengths similar to those described above for leg **318**.sup.f, in which case, some portions of clinch portions **328** of both legs, including the radial clinch sections **336** and/or axial clinch sections **334**, may be configured to at least partially overlap over each other in the second bent configuration.

[0134] FIG. **17** shows an exemplary prosthetic valve **100**.sup.c. Prosthetic valve **100**.sup.c is an exemplary implementation of prosthetic valve **100**, and thus includes all of the features described for prosthetic valve **100** throughout the current disclosure, except that the leaflets **162** are coupled to the frame **110**.sup.c, for example along their cusp end portions **165**, via stapling members **300**. The frame **110**.sup.c shown in FIG. **17** can be generally similar to the frame **110**.sup.b, comprising a plurality of struts **114**, such as angled struts **115** that include a group of first struts **115***a* to which the leaflets **162** are not coupled along their cusp end portions **165**, and a group of second struts **115***b* to which the cusp end portions **165** are coupled, forming a scalloped line **105** along selected angled struts **115***b*.

[0135] Any number of stapling members **300** can be utilized. In some examples, a single stapling member **300** can be utilized to couple all leaflets **162** to the frame **110**. For example, a single stapling member **300** can have an undulating or scalloped-shaped extending across the complete circumference of the frame **110**. In some examples, the number of stapling members **300** can be similar to the number of the leaflets **162**, For example, for a tri-leaflet valvular structure **160**.sup.c, three cusp-shaped stapling members **300** can be used, each formed to generally follow the shape of the cusp of one of the leaflets **162**. In some examples, the number of the stapling members **300** can be twice the number of leaflets **162**. For example, for a tri-leaflet valvular structure **160**.sup.c, six cusp-shaped stapling members **300** can be used, each shaped to extend between an inflow apex **129** and a commissure **180**, such that each stapling member **300** is used to couple about half of the cusp end portion **165**. In some examples, several stapling members **300** can be used to couple different portions of the cusp edge **164** to the frame **110**. For example, separate stapling members **300** can be used to extend over each strut **115***a* separately.

[0136] The shape of the stapling members **300** can generally follow the shape of the struts **115** to which they are coupled. For example, the second struts **115***b* illustrated in FIG. **17** are shown to extend in a zig-zagged pattern between each inflow apex **129** and a commissure **180**, due to the junctions **127** disposed between adjacent struts **115***b*, causing each second strut **115***b* to be offset relative to the neighboring second strut **115***b*. A stapling member **300** configured to extend along more than one strut **115***b*, can be shaped to exhibit a similar zig-zagged pattern.

[0137] FIG. **33** shows a portion of an exemplary stapling member **3008**. Stapling member **300**.sup.g is an exemplary implementation of stapling member **300**, and thus includes all of the features described for stapling member **300** throughout the current disclosure, except that the spine **3028** extends in a zig-zagged pattern, shaped to generally match the zig-zagged pattern of the second struts **115***b* shown in FIG. **17** for example. In some examples, intermediate portion **316** of spine **3028** includes an offsetting portion **338** disposed between sub-portions **320**, wherein the sub-portions **320** can be parallel to each other, yet offset from each other in a direction which is angled relative to the orientation of the sub-portions **320**. That is to say, the stapling member axis Cs extends in a zig-zagged pattern, such that the offsetting portions **338** are angled relative to the sub-portions **320**. When used to coupled leaflets **162** to the frame **102**, the offsetting portions **338** are aligned with corresponding junctions **127**, and are shaped and sized to align the sub-portions **320** with the angled struts **115** extending from the junctions **127**.

[0138] In some examples, the legs **318** of stapling portion **310** can have different leg lengths L.sub.L. Stapling portion **3108** is illustrated in FIG. **33** to include a first leg **3188**.sup.g having a leg length LLa and a second leg **3188**.sup.g having a leg length L.sub.Lb, wherein the length of the first leg **318**.sup.ga is greater than the length of the second leg **3188**.sup.g (i.e., L.sub.La>L.sub.Lb). While the legs **318**.sup.ga and **318**.sup.g*b* are illustrated to be aligned with each other in FIG. **33**, it is to be understood that this is shown by illustration and not limitation, and that in some examples, legs **318** having unequal leg lengths can be offset from each other, such as shown and described for stapling members **300**.sup.d for example.

[0139] While pairs of legs having unequal lengths is shown in combination with a zig-zagged spine in FIG. **33**, it is to be understood that these features can be used in isolation, and combined with features of any other exemplary stapling member disclosed herein. For example, a zig-zagged spine, such as spine **3028**, can be used in combination with stapling portions **310** equipped with legs **318** having equal lengths (see FIGS. **5**A-**16**), which can be aligned (see FIGS. **5**A-**12**) or offset legs (see FIGS. **13-16**), or stapling members **310** equipped with a single leg **318** (see FIGS. **32**A-**32**B). Similarly, legs having unequal lengths, such as legs **3188**, can be implemented with spines which are not necessarily zig-zagged, such as curved spine **302**.sup.a for example.

[0140] Referring back to FIG. **17**, While prosthetic valve **100**.sup.c is shown without an inner skirt **106** or an outer skirt **107**, it is to be understood that any of an inner skirt **106** and/or an outer skirt **107** can be added, in which case, the length L.sub.R of the radial portions **326** of the legs **318** can be adapted to extend through the thicknesses of the inner skirt **106** and/or outer skirt **107** as well. When an outer skirt **107** is added, the legs **318** can either penetrate through the outer skirt **107** or not. In some examples, the legs **318** extend through the thickness of an outer skirt **107** extending around the frame **110**, such that the clinch portions **328** are bent over an outer surface of the outer skirt **107**. In some examples, the legs **318** are bent over the struts **115**, such that the clinch portions **328** extend over strut outer surfaces **113**, and the outer skirt **107** is then attached to the frame **110** around and over the clinch portions **328**. Such examples can advantageously conceal the tips **330** from contacting the surrounding anatomy around an implanted prosthetic valve **100**.

[0141] In some examples, the leaflets include increased-stiffness portions through which stapling members can extend to couple the valvular structure to the frame. FIG. **18** shows an example of a leaflet **162**.sup.e. Leaflet **162**.sup.e is an exemplary implementation of leaflet **162**, and thus includes all of the features described for leaflet **162** throughout the current disclosure, except that leaflet **162**.sup.e further comprises a stiff portion **172** with a stiffness that is greater than the stiffness of the leaflet body **170**. The stiff portion **172** can extend from the cusp edge **164** and terminate at a stiff portion proximal end **174**, which can generally track the shape of the cusp edge **164**. The stiff portion proximal end **174** can be the border between the leaflet body **170**, which is the movable portion of leaflet **162**.sup.e, and the stiff portion **172**, defining stiff portion width W.sub.SP between the cusp edge **164** and the stiff portion proximal end **174**. The stiff portion **172** can serve as the cusp end portion **165**.sup.e of leaflet **162**.sup.e.

[0142] The stiff portion **172** can extend all the way up to the tabs **168**, or it can terminate below the level of the tabs, as illustrated in FIG. **18**. In some examples, the stiff portion proximal end **174** is parallel to the cusp edge **164**, such that the stiff portion width W.sub.SP is uniform along the stiff portion **172**. In other examples, the stiff portion width W.sub.SP can vary, for example from a wider width at the lower tip of the leaflet to a narrower width at the upper ends of the stiff portion **172**, closer to tabs **168**.

[0143] Leaflet **162**.sup.e can be formed, in some examples, from a unitary continuous piece of material, wherein different portions of the leaflet **162**.sup.e can be treated to have different material properties. Leaflet **162**.sup.e can be formed from natural tissue, such as bovine pericardium, undergoing biological treatment procedures that can include subjecting the tissue to cross-linking agents, which can influence its final material properties. In such an example, each of stiff portion **172** can be subjected to a different biological procedure than the remainder of the leaflet, including leaflet body **170** and optionally tabs **168**, such as: being subjected to different cross-linking agents, subjected to such agents for different time durations, or other procedural variations, adapted to result in stiff portion **172** that can be stiffer than the leaflet body **170**. Thus, the leaflet body **170** of leaflet **162**.sup.e, which has material properties comparable to those of the leaflet body **170** of conventional leaflets **162** (including, for example, leaflet **162**.sup.a), is the portion of the leaflet **162**.sup.e which is not directly attached to the frame **110**, but is rather configured to freely move toward the frame **110** in an open state of the valvular structure **160**.sup.e, and toward valve central longitudinal axis Ca in a closed state of the valvular structure **160**.sup.e.

[0144] As mentioned, leaflet **162**.sup.e can optionally exhibit a different degree of cross-linking in different portions thereof, such that the stiff portion **172** can be more highly cross-linked than the leaflet body **170**. This can be achieved, in some implementations, by cross-linking the stiff portion **172** with a first cross-linking agent or solution and cross-linking the leaflet body **170** with a second cross-linking agent. Alternatively or additionally, the stiff portion **172** can be cross-linked during a longer period of time relative to the leaflet body **170**. Cross-linking agents can include, but are not limited to, divinyl sulfone (DVS), polyethylene glycol divinyl sulfone (VS-PEG-VS), hydroxyethyl methacrylate divinyl sulfone (HEMA-DIS-HEMA), formaldehyde, glutaraldehyde, aldehydes, isocyanates, alkyl and aryl halides, imidoesters, N-substituted maleimides, acylating compounds, carbodiimide, hexamethylene diisocyanate, 1-Ethyl-3-[3-dimethylaminopropyl] carbodiimide hydrochloride (EDC or EDAC), hydroxychloride, N-hydroxysuccinimide, and combinations thereof.

[0145] The leaflet body **170** can be designed to have material properties substantially similar to those of conventional leaflets **162** described above with respect to FIGS. **1**A-**2**, allowing this portion of the leaflet to move between the closed and open states in a similar manner. The stiff portion **172** can be stiffer compared to the leaflet body **170**, facilitating easier penetration of stapling members **300** utilized to couple these portions to the frame during valve assembly, and improving engagement retention with the stapling members **300**. The stiff portion **172** can be semi-rigid, meaning that it is stiffer or more rigid than the leaflet body **170**, yet flexible enough to transition between the crimped and expanded configurations of the prosthetic valve without experiencing material failure and without resisting such transitions of the valve.

[0146] In some examples, the ultimate tensile stress of the stiff portion **172** is at least 1.5 times greater than the ultimate tensile stress of the leaflet body **170**. In some examples, the ultimate tensile stress of the stiff portion **172** is at least 2 times greater than the ultimate tensile stress of the leaflet body **170**. For example, for a leaflet body **170** having an ultimate tensile stress of about 1 MPa (Megapascal), stiff portion **172** can have an ultimate tensile stress which is greater than 1.5 MPa, and in some cases, greater than 2 MPa.

[0147] In some examples, the stiff portion **172** obtains a load at failure which is at least 2 times greater than the load at failure obtained by the leaflet body **170**. In some examples, the stiff portion **172** obtains a load at failure which is at least 3 times greater than the load at failure obtained by the leaflet body **170**. For example, for a leaflet body **170** that obtains a load at failure of about 3 N (Newton), the stiff portion **172** can obtain a load at failure which is greater than 6 N, and in some cases, greater than 9 N.

[0148] The stiffness of the stiff portion **172** can be homogenous or non-homogenous. For example, the stiffness of the stiff portion **172** can be homogenous between the cusp edge **164** and the stiff portion proximal end **174**, or it can vary from higher stiffness at the cusp edge to lower stiffness at the stiff portion proximal end **174**. In such cases, the minimal stiffness of the stiff portion **172** will be equal to or greater than the stiffness of the leaflet body **170**.

[0149] In some examples, the entire leaflet **162**.sup.e can be provided as a unitary continuous material, wherein the stiff portion **172** can be reinforced by coating with a reinforcing layer, or otherwise attached along a surface thereof to a reinforcing layer that imparts additional stiffness thereto, such as a polymeric layer or any other suitable layer. The stiffness of any portion of a valvular structure can be measured by following the general procedures set forth in ASTM D790.

[0150] In some examples, the leaflet **162**.sup.e can be formed from two different materials, joined to each other along stiff portion proximal end **174**, which serves in such cases as an attachment line **174** (achieved, for example, by suturing, gluing, welding, and the like), wherein the material forming the stiff portion **172** can be stiffer than the material forming the leaflet body **170**.

[0151] FIG. **19** shows a portion of a leaflet **162**.sup.e coupled to a strut **114** by stapling member **300**. Attachment of leaflet **162**.sup.e to strut **114** can be performed according to any of the examples described above with respect to FIGS. **10-12**, **15**A-**16** and **31-32**B, utilizing any of the exemplary stapling members **300** described above with respect to FIGS. **5**A-**9**B, **13**A-**14**B and **31-33**, wherein the stiff portion **172** of the leaflet **162**.sup.e is positioned adjacent to strut **114**, and the legs **318** pierce and extend through the stiff portion **172**. Advantageously, the increased stiffness of the stiff portion **172** allows it to better withstand stresses concentrated at the points of leg **318** penetration.

[0152] In some examples, the stiff portion width W.sub.SP is greater than the strut width W.sub.S. In some examples, the stiff portion width W.sub.SP is at least 1.5 times greater than the strut width W.sub.S. In some examples, the stiff portion width W.sub.SP is at least 2 times greater than the strut width W.sub.S. In some examples, the stiff portion width W.sub.SP is greater than the backspan width W.sub.B of backspan **314**. In some examples, the stiff portion width W.sub.SP is at least 1.5 times greater than the backspan width W.sub.B. In some examples, the stiff portion width W.sub.SP is at least 2 times greater than the backspan width W.sub.B.

[0153] In some examples, a stapling member can be formed from a continuous wire. FIGS. **20**A-**20**C show a portion of an exemplary stapling member **350** that can be used for coupling a leaflet **162** to one or more struts **114** of a frame **110**. FIG. **20**A shows the stapling member **350** in a flattened configuration. Stapling member **350** is formed from a continuous wire **351** that can be bent to form a spine **352** which can generally extend along stapling member axis Cs in the same manner described above for stapling member **300** and illustrated in FIG. **5**A, and a plurality of stapling portions **360** separated from each other by intermediate wire portions **366** of the spine **352**. The wire **351** can be made of plastically-deformable materials such as, but not limited to, stainless steel or a nickel based alloy (e.g., a cobalt-chromium or a nickel-cobalt-chromium alloy such as MP35N alloy), titanium, and the like.

[0154] The wire **351** of each stapling portion **360** is bent to define two legs **368** extending from the opposite sides of the spine, such as a first leg **368***a* and a second leg **368**.sup.b. Each leg **368** terminates at a tip **380**, which can be, in some examples, a relatively sharp tip configured to penetrate through the thickness of a leaflet **162**, such as through a pericardial tissue the leaflet **162** can be formed of, and in some examples can be a blunt tip that can be passed through pre-formed perforations **176** of a leaflet **1624**, in a similar manner to that described above with respect to FIG. **31**. Formation of a stapling portion **310** can be achieved by laterally bending the wire **351** from and end of a corresponding intermediate wire portion **366**, sideways (relative to the stapling member axis Cs), for example at a right angle (though other angles are contemplated) in a first direction, so as to form a first planar bent extension **382**, up to a first tip **380***a*. The wire **351** is bent to form a second planar bent extension **384** that extends in an opposite direction from the first tip **380***a* up to a second tip **380**.sup.b. The wire **351** finally is bent to form a third planar bent extension **386** that extends from the second tip **380**.sup.b back (i.e., in the first direction) to the starting point of a subsequent intermediate wire portion **366**, and is bent, for example at a right angle (though other angles are contemplated) to form the subsequent intermediate wire portion **366**. The length of each of the first planar bent extension **382** and third planar bent extension **386** can be similar, is defined as the leg length L.sub.L in a similar manner to that described above for a leg length of stapling members **300**.

[0155] In the flattened configuration shown in FIG. **20**A, the first leg **368***a* can be defined by the first planar bent extension **382** and a portion of the second planar bent extension **384** that extends next to the first planar bent extension **382** along the leg length L.sub.L. The second leg **368**.sup.b can be defined by the third planar bent extension **386** and a portion of the second planar bent extension **384** that extends next to the third planar bent extension **386** along the leg length L.sub.L.

[0156] In some examples, the tips **380** are formed by relatively linearly lateral bent extensions of the wire **351**. FIGS. **20**A-**20**C show an exemplary stapling member **350**.sup.a, which is an exemplary implementation of stapling member **350**, and thus includes all of the features described for stapling member **350** throughout the current disclosure, except that the tips **380**.sup.a are formed by relatively linearly lateral bent extensions of the wire **351**. Specifically, the first tip **380**.sup.a a is formed by bending the first planar bent extension **382**.sup.a, for example at a right angle (though other angles are contemplated) to form a first lateral bent extension **388**, which is then bent again, for example at a right angle (though other angles are contemplated) to form the second planar bent extension **384**.sup.a. Similarly, the second tip **380**.sup.a*b* is formed by bending the second planar bent extension **384**.sup.a, for example at a right angle (though other angles are contemplated) to form a second lateral bent extension **390**, which is then bent again, for example at a right angle (though other angles are contemplated) to form the third planar bent extension **386**.sup.a.

[0157] The first and second lateral bent extensions **388**, **390** can be thin enough or sharpened to allow convenient penetration of the tips **380**.sup.a through the leaflet material. The lateral tip length L.sub.T of each of the first lateral bent extension **388** extending between the first planar bent extension **382**.sup.a and second planar bent extension **384**.sup.a, and the second lateral bent extension **390** extending between the second planar bent extension **384**.sup.a and third planar bent extension **386**.sup.a, can be substantially less than the leg length L.sub.L, to facilitate penetration of the tips through the leaflet material. In some examples, the lateral tip length L.sub.T is less than one fifth of the leg length L.sub.L (i.e., L.sub.T<0.2 L.sub.L). In some examples, the lateral tip length L.sub.T is less than one tenth of the leg length L.sub.L (i.e., L.sub.T<0.1 L.sub.L).

[0158] FIG. **20**B shows the stapling member **350** in a first bent configuration, wherein the legs **368** are bent at first bends **372**, for example at right angles (though other angles are contemplated) in a radial direction (i.e., out of plane relative to the plane defined by the stapling member **350** in the planar configuration illustrated in FIG. **20**A), such that each stapling portion **360** can be generally U-shaped. The portions of the first, second, and third planar bent extensions **382**, **384**, **386**, disposed between the first bends **372** of both legs **368***a*, **368**.sup.b, together form the backspan **364** of stapling portion **360**, configured to function in a similar manner described above for backspan **314**, pressing against the inner surface of the leaflet **162** when coupling the leaflet **162** to a strut **114**. The distance between the first bends **372** of both legs **368***a*, **368**.sup.b is defined as the backspan width W.sub.B in a similar manner to that described above for a backspan width of stapling members **300**.
[0159] FIG. **20**C shows the stapling member **350** in a second bent configuration, wherein the legs **368** are further bent towards each other at second bends **374**. As shown, each leg is bent at a second bend **374**, between the first bend **372** and the tip **380**, dividing the leg **368** into a radial portion **376** extending between the first bend **372** and the second bend **374**, and a clinch portion **378** extending between the second bend **374** and the tip **380**. The radial portion **376** can extend along a radial portion length L.sub.R, and the clinch portion **378** can extend along a clinch portion length L.sub.C, in a similar manner to that described above for a radial portion length and a clinch portion length of stapling members **300**. Since a portion of the leg **318** also defines part of the backspan **364**, up to the first bend **372**, the sum of lengths of the remaining radial portion **376** and clinch portion **378** can be less than the leg length L.sub.L (i.e., L.sub.R+L.sub.C<L.sub.L).
[0160] The first and second clinch portions **378***a*, **378***b* extend towards each other, and can be generally parallel to the backspan **364**. As shown, the legs **368** of each stapling portion **360** formed in the manner described above, can be offset from each other, such that when the legs **368** are bent at second bends **374**, the clinch portion **378** do not intersect with each other but rather extend parallel to each other. As described above with respect to stapling member **3004**, this offset configuration allows the clinch portion **378** to be longer than half the backspan width W.sub.B, without the risk of the tips **380** contacting or otherwise interfering with each other. This, in turn, can increase surface contact area between the clinch portions **378** and strut **114**, to improve clamping force.
[0161] FIGS. **21**A-**21**C show an exemplary stapling member **350**.sup.b, illustrated in a flattened configuration (FIG. **21**A), a first bent configuration (FIG. **22**B), and a second bent configuration (FIG. **21**C). Stapling member **350**.sup.b is an exemplary implementation of stapling member **350**, and thus includes all of the features described for stapling member **350** throughout the current disclosure, except that the tips **380**.sup.b are pointed or in the shape of an arrowhead. Each leg **368**.sup.b can be generally triangular in the flattened configuration shown in FIG. **21**A. The first planar bent extension **382**.sup.b and the second planar bent extension **384**.sup.b define an angle α therebetween, and the second planar bent extension **384**.sup.b and the third planar bent extension **386**.sup.b similarly define the angle α therebetween. In some examples, the angle α is an acute angle. In some examples, the angle α is less than 20°. In some examples, the angle α is less than 10°. In some examples, the angle α is less than 5°.
[0162] Utilization of a stapling member **350** for coupling a leaflet **162** to strut **114** can be performed in the same manner described above for stapling members **300** with respect to FIGS. **10**-**12** and **15**A-**16**, mutatis mutandis. For example, a cusp end portion **165** of a leaflet **162** can be placed adjacent to one or more struts **114** of the frame **110**, such that the cusp edge **164** can be somewhat lower than the strut **114**, as illustrated in FIG. **10**. The leaflet is placed against the inner surface of the frame **110**, such that the leaflet **162** is facing the strut inner surface **112**. Stapling member **350**, in a first bent configuration thereof, can be then approximated to the leaflet **162** from the inner side, such that the legs **368** are oriented toward the leaflet **162**. The legs **368** are then pressed into the leaflet **162**, causing the tips **380** to pierce through the tissue material, and the legs

368 penetrate into leaflet **162** and extend outward beyond the strut outer surface **113**, sandwiching the leaflet **162** between the backspan **364** and the strut **114**. Legs **368** are then bent over the outer surface **113** of the strut **114**, transitioning the stapling member **350** to the second bent configuration, thus clamping the stapling member **350** over strut **114** to couple the leaflet **162** to the frame **110**, for example through and along cusp end portion **165**.

[0163] In some implementations, the stapling member **350** can be generally curved, meaning that it extends along a curved stapling member axis Cs, optionally following a curvature of the cusp edge **164** and/or cusp end portion **165** of leaflet **162**. In some implementation, the curved shape of one or more stapling member **350** can follow a scalloped shape of the scalloped line **105**, to allow coupling of the valvular structure **160**, by one or more stapling members **300**, along a generally scalloped-shaped line **105**. Intermediate portion **366** can extend along an intermediate portion length L.sub.I (see FIG. **20**A) between adjacent stapling portions **360**, similar to an intermediate portion length described above with respect to stapling members **300**. Different intermediate portions **366** of the spine **352** can have similar or different lengths L.sub.I. For example, intermediate portions **366** extending between subsequent stapling portion **310** configured to couple a leaflet **162** to a section of a strut **114** extending between two neighboring junctions **127** can be shorter in length than intermediate portions **316** that extend along junctions **127** of the frame. In some examples, spine **352** can extend in a generally zig-zagged pattern, in a similar manner to that described above with respect to spine **3028** of FIG. **33**. While not illustrated explicitly, intermediate wire portions **366** can have offsetting portions angled relative to sub-portions extending from the offsetting portions, similar to offsetting portions **338** and sub-portion **320** described with respect to FIG. **33**.

[0164] Radial portions **376** of legs **368** extend through the thickness of the leaflet **162** and across the strut **114**. In some examples, the radial portion length L.sub.R is substantially equal to the combined thicknesses of the leaflet **162** and the strut **114** (i.e., L.sub.R≈T.sub.L+T.sub.S). In some examples, the radial portion length L.sub.R is within the range of ±20% of the combined thicknesses of the leaflet **162** and the strut **114**, such that 0.8 (T.sub.L+T.sub.S)<L.sub.R<1.2 (T.sub.L+T.sub.S).

[0165] As mentioned above, the directions of attachment, extending the legs **368** from the inner side of the frame **110** outwards, may be advantageous, resulting in the relatively sharp tips **380** disposed on the outer side of the valve **100**, which reduces the risk of these tips **380** being otherwise engaged by movable portions of the leaflets **162** inside the valve **100**.

[0166] One or more stapling members **350** can be used to couple leaflets **162** to the struts **114** of a frame **110**, in the same manner described for any of the examples above with respect to FIG. **17**, mutatis mutandis. For example, any number of stapling members **350** can be utilized. In some examples, a single stapling member **350** can be utilized to couple all leaflets **162** to the frame **110**. For example, a single stapling member **350** can have an undulating or scalloped-shaped extending across the complete circumference of the frame **110**. In some examples, the number of stapling members **350** can be similar to the number of the leaflets **162**, For example, for a tri-leaflet valvular structure **160**.sup.c shown in FIG. **17**, three cusp-shaped stapling members **350** can be used, each formed to generally follow the shape of the cusp of one of the leaflets **162**. In some examples, the number of the stapling members **350** can be twice the number of leaflets **162**. For example, for a tri-leaflet valvular structure **160**.sup.c shown in FIG. **17**, six cusp-shaped stapling members **350** can be used, each shaped to extend between an inflow apex **129** and a commissure **180**, such that each stapling member **350** is used to couple about half of the cusp end portion **165**. In some examples, several stapling members **350** can be used to couple different portions of the cusp edge **164** to the frame **110**. For example, separate stapling members **350** can be used to extend over each strut **115***a* (shown in FIG. **17**) separately.

[0167] The shape of the stapling members **350** can generally follow the shape of the struts **115** to which they are coupled. For example, the second struts **115***b* illustrated in FIG. **17** are shown to extend in a zig-zagged pattern between each inflow apex **129** and a commissure **180**, due to the junctions **127** disposed between adjacent second struts **115***b*, causing each strut **115***b* to be offset relative to the neighboring strut **115***b*. A stapling member **350** configured to extend along more than one strut **115***b*, can be shaped to exhibit a similar zig-zagged pattern.

[0168] In some examples, any of an inner skirt **106** and/or an outer skirt **107** can be added, in which case, the length L.sub.R of the radial portions **376** of the legs **368** can be adapted to extend through the thicknesses of the inner skirt **106** and/or outer skirt **107** as well. When an outer skirt **107** is added, the legs **368** can either penetrate through the outer skirt **107** or not. In some examples, the legs **368** extend through the thickness of an outer skirt **107** extending around the frame **110**, such that the clinch portions **378** are bent over an outer surface of the outer skirt **107**. In some examples, the legs **368** are bent over the struts **115**, such that the clinch portions **378** extend over strut outer surfaces **113**, and the outer skirt **107** is then attached to the frame **110** around and over the clinch portions **378**. Such examples can advantageously conceal the tips **380** from contacting the surrounding anatomy around an implanted prosthetic valve **100**.

[0169] In some examples, stapling members **350** are used to couple leaflets **162**.sup.e that includes a stiff portion **172** to the frame **110**, in a manner similar to that described above with respect to FIGS. **18-19**, mutatis mutandis. In such examples, the legs **368** pierce and extend through the stiff portion **172**, wherein the increased stiffness of the stiff portion **172** advantageously allows it to better withstand stresses concentrated at the points of leg **368** penetration. In some examples, the stiff portion width W.sub.SP is greater than the backspan width W.sub.B of backspan **364**. In some examples, the stiff portion width W.sub.SP is at least 1.5 times greater than the backspan width W.sub.B. In some examples, the stiff portion width W.sub.SP is at least 2 times greater than the backspan width W.sub.B.

[0170] In some examples, compressible inserts can be utilized to clamp leaflets **162** to a frame **110** that includes corresponding recesses into which the compressible inserts can be inserted. FIG. **22** shows a portion of an exemplary frame **110**.sup.d of a prosthetic valve **100**.sup.d. Prosthetic valve **100**.sup.d is an exemplary implementation of prosthetic valve **100**, and thus includes all of the features described for prosthetic valve **100** throughout the current disclosure, except that at least some of the struts **114** of frame **110**.sup.d further include recesses **120** with openings **122** facing the valve central longitudinal axis Ca. In some examples, the frame **110**.sup.d can comprise a plurality of struts **114**, such as angled struts **115** that include, similar to frames **110**.sup.b and **110**.sup.c described above, a group of first struts **115***a* to which the leaflets **162** are not coupled along their cusp end portions **165**, and a group of second struts **115***b* to which the cusp end portions **165** are coupled, forming a scalloped line **105** along selected angled struts **115***b*.

[0171] At least some of the struts **114** of frame **1104**, which can be second struts **115***b* to which leaflets **162** are to be coupled, such as along their cusp end portions **165**, include recesses **120**. A recess **120**, also shown in cross-section in FIG. **25**A, includes an opening **122** formed at the strut inner surface **112**, and terminates radially inward to strut outer surface **113**. Opening **122** defines an opening width W.sub.O. While recess **120** is shown to be circular in FIG. **25**A, it is to be understood that other shapes, such as elliptical, rectangular, and the like, are contemplated. The recess **120** has a maximal recess width W.sub.R in a direction parallel to the opening width W.sub.O, and a recess depth H.sub.R extending from the opening **122** to an opposite floor of the recess, in a radial direction, parallel to strut thickness T.sub.S. In some implementations, when the recess is substantially circular, the maximal recess width W.sub.R can be its internal diameter, and the recess depth H.sub.R can be optionally also equal to its diameter. The maximal recess width W.sub.R is greater than the width W.sub.O of opening **122**, such that opening **122** extends between two shoulders **124** formed by the strut **114** along the strut inner surface **112**. The recess depth H.sub.R is smaller than the strut thickness T.sub.S.

[0172] Struts **114** that include recesses **120** can be formed to have first strut widths Ws′ which are relatively wider than those of equivalent struts **114** of conventional frames **110**, devoid of such

recesses. Struts **114** that include recesses **120** can be also formed to have first strut thicknesses Ts′ which are relatively thicker than those of equivalent struts **114** of conventional frames **110**. In some examples, as illustrated for frame **110**.sup.d in FIG. **22**, angled struts **115***b* to which the leaflets are coupled, as well as struts **115***a* to which no leaflets are coupled, can be provided with similar first strut widths Ws′ and/or similar first strut thicknesses Ts′, which are wider and/or thicker, respectively, than those of equivalent struts **114** of conventional frames **110**. In some examples, a frame **110**.sup.d can include a third group of angled struts **115***c*, such as struts **115** diverging from outflow apices **128**. These struts **115***c* can be disposed at an outflow portion of the frame **110**.sup.d which is proximal to the valvular structure **160**, and more specifically, proximal to the regions of attachment of the leaflets **162**, along cusp end portions **165** thereof, to the frame **110**.sup.d. These struts **115***c* can be optionally formed with second strut widths Ws″ that can be generally similar to the strut width of conventional frames **110**, such that the second strut width Ws″ of outflow struts **115***c* can be smaller than the first strut width Ws′ of struts **115***b* that include recesses **120**.
[0173] FIG. **23** shows a portion of an exemplary frame **110**.sup.e of a prosthetic valve **100**.sup.e. Prosthetic valve **100**.sup.e is an exemplary implementation of prosthetic valve **100**.sup.d, and thus includes all of the features described for prosthetic valve **100**.sup.d throughout the current disclosure, except that only the angled struts **115***b* that include recesses **120** are wider, having the first strut widths Ws′, while other angled struts, such as angled struts **115***a*, and optionally outflow struts **115***c*, can remain thinner, having second strut widths Ws″ generally similar to those of equivalent struts **114** of conventional frames **110**. In some examples, the first strut width Ws′ is at least 1.5 times greater than the second strut width Ws″. In some examples, the first strut width Ws” is at least 2 times greater than the second strut width Ws″.
[0174] FIG. **24** shows an exemplary compressible insert **340**, having an insert diameter D.sub.I in a free state thereof. The insert is squeezable to a size which is less than its free state diameter D.sub.I, and is configured to be inserted into a corresponding recess **120**. Once inside the recess **120**, the insert **340** strives to revert back to its free inset diameter D.sub.I. The compressible insert **340** can be made from any suitable resilient squeezable material, such as rubber or other polymeric material.
[0175] FIGS. **25**A-**25**B shows cross-sectional views of a strut **114** comprising a recess **120**, prior to (FIG. **25**A) and after (FIG. **25**B) coupling a leaflet **162** to the strut **114** with a compressible insert **340**. The strut **114** that includes a recess **120** defines a first strut width Ws′ and a first strut thickness Ts′. The maximal recess width W.sub.R is smaller than the first strut width Ws′. The recess depth H.sub.R is smaller than the first strut thickness Ts′.
[0176] As shown in FIG. **25**A, a leaflet **162** is approximated to the strut inner surface **112**. The compressible insert **340** is pushed from the opposite side of the leaflet **162**, so as to push it into the recess **120**, as shown in FIG. **25**B. In some examples, the insert diameter D.sub.I in its free state (shown in FIG. **25**A) is greater than the opening width W.sub.O, forcing the compressible insert **340** to be squeezed to a diameter which is less than the opening width W.sub.O as it is pushed through the opening **122**. Since the compressible insert **340** folds the leaflet **162** thereover and drags the folded portion of the leaflet **162** therewith through opening **122**, it needs to be able to be squeezed into a width which is less than the opening width W.sub.O minus twice the leaflet thickness T.sub.L, accounting for the two layers on both sides of the insert **340** dragged therewith through the opening **122**. In some examples, the insert diameter D.sub.I in its free state does not necessarily needs to be greater than the opening width W.sub.O, but rather at least greater than the gap G.sub.O between the two layers of leaflet **162** extending through the opening **122** (see FIG. **25**B), such that G.sub.O=W.sub.O−2T.sub.L, and D.sub.I>Go, but is squeezable to a size of the gap G.sub.O when forces to pass through the opening **122**.
[0177] Once positioned within the recess **120**, as shown in FIG. **25**B, the insert **340** can expand to a diameter which is greater than the gap G.sub.O, which can be its free state diameter D.sub.I or a diameter which is less than D.sub.I (but still greater than G.sub.O), if full expansion of the insert

340 to its free state diameter D.sub.I is limited by the folded leaflet layers disposed on both sides of the insert **340**, between the insert **340** and the inner walls of the recess **120**. As further shown in FIG. **25**B, the leaflet in this final configuration extends around both shoulders **124**, wherein the insert **340** disposed within the recess **120** is pressing the leaflet portions against the shoulders **124**, such that the shoulders **124** serve as ridges that prevent the leaflet **162** and insert **340** from spontaneously sliding out of the recess **120**. The effectively snap-fits the leaflet **162** into recess **120**. In some examples, the insert diameter D.sub.I is smaller than the recess depth H.sub.R. In some examples, the sum of insert diameter D.sub.I and twice the leaflet thickness T.sub.L is smaller than the recess depth H.sub.R, such that D.sub.I<(H.sub.R-2T.sub.L).

[0178] In some examples, a plurality of struts **114**, such as the plurality of angled second struts **115***b*, can include a corresponding plurality of recesses **120**. For example, one recess **120** can extend between the junctions **127** (or a junction **127** and an inflow apex **129**) of each corresponding strut **114** to which the leaflet **162** is to be coupled. As shown in FIGS. **22-23**. The struts **115***b* that include recesses **120** can be struts that generally follow a scalloped-shaped line. In some examples, the length of the plurality of recesses **120** can be non-equal, for example, when formed into struts **114** having non-equal lengths. While each strut **115***b* is shown in FIGS. **22-23** to include a single recess, it is to be understood that in some example, more than one recess **120** can be formed in a single strut **114**. While adjacent recesses **120** are shown in FIGS. **22-23** to be separated from each other by junctions **127**, it is to be understood that in some examples, a single recess can span across more than one strut **114**, formed in a junction **127** between the struts **114** as well.

[0179] The length of the compressible insert **340** can be similar to or less than the length of the corresponding recess **120**. In some examples, a single compressible insert **340** is inserted into a corresponding one of the recesses **120** to couple the leaflet **162** to the strut **114**. In some examples, a plurality of compressible insert **340**, each of which is shorter in length than the corresponding recesses **120**, are inserted into the corresponding recess **120** to couple the leaflet **162** to the strut **114**.

[0180] Strut **114** that includes a recess, are also referred to as “recessed struts”. Thus, the terms “recessed struts” and “second struts **115***b*”, for example with reference to prosthetic valves such as **100**.sup.d or **100**.sup.e, are interchangeable. Similarly, struts **114** which are devoid of recesses **120**, are also referred to as “unrecessed struts”. Thus, the term “unrecessed struts” can be used interchangeably with any of “first struts **115***a*” and/or “third struts **115***c*”, for example with reference to prosthetic valves such as **100**.sup.d or **100**.sup.e. The cross-sectional view of FIGS. **25**A-**25**B are taken across recessed struts. While angled struts **115***b* are described above and illustrated in FIGS. **22-23** as recessed struts, it is to be understood that this is not meant to be limiting, and that recessed struts can similarly include axial struts. While leaflets **162** are described above to be coupled to recessed struts along cusp end portions **165** thereof, this is not meant to be limiting, and other portions of the leaflets, such as those including tabs or equivalent regions extending between the cusp edges **164** and the free edges **166** can be also coupled to recesses formed, for example, in vertical struts, thereby coupling the leaflets to the frame at commissure regions as well.

[0181] FIG. **26** shows an example of a conventional surgically implantable prosthetic valve **500**. In the particular illustrated example, the prosthetic valve **500** comprises a support frame **510** and a valvular structure **560** attached thereto. The valvular structure **560** comprises a plurality of leaflets **562** (e.g., three leaflets), positioned at least partially within the frame **510**, and configured to regulate flow of blood through the prosthetic valve **500**. While three leaflets **562** arranged to collapse in a tricuspid arrangement similar to the native aortic valve, are shown in the example illustrated in FIG. **26**, it will be clear that a prosthetic valve **500** can include any other number of leaflets **562**, such as two leaflets configured to collapse in a bicuspid arrangement similar to the native mitral valve, or more than three leaflets, depending upon the particular application. The leaflets **562** can be generally similar to leaflets **162** described above, and can be made of a flexible material, derived from biological materials (e.g., bovine pericardium or pericardium from other sources), bio-compatible synthetic materials, or other suitable materials as known in the art and described, for example, in U.S. Pat. Nos. 6,730,118, 6,767,362 and 6,908,481, which are incorporated by reference herein.

[0182] The frame **510** comprises base portion **512** which is generally rigid and/or expansion-resistant in order to maintain the particular shape and diameter of the prosthetic valve **500**, and a plurality of vertically oriented commissure posts **518** (e.g., three posts) extending proximally from the base portion **512** to support the free edges **566** of the leaflets **562**. The base portion **512** can comprise cusp portions **516** extending between the vertically oriented commissure posts **518**. The frame **510** can be metallic, plastic, or a combination of the two.

[0183] Each of the leaflets **562** can be attached along a cusp edge **564** thereof to a corresponding cusp portion **516** of the frame **510** and up along adjacent commissure posts **518**. Each leaflet **562** can include a pair of oppositely-directed tabs **568**, wherein each respective tab **568** can be aligned with a tab **568** of an adjacent leaflet **562** as shown. The tabs **568** can be wrapped around a respective commissure post **518** of the frame **510** and coupled thereto, thereby forming commissures **580** that project in an outflow direction along the valve central longitudinal axis Ca.

[0184] A soft sealing or sewing ring **522** circumscribes the frame **510**, for example around the base portion **512**, and is typically used to secure the prosthetic valve to a native annulus such as with sutures. The sewing ring **522** comprises a sewing ring insert **524** and a cloth cover **526**. The sewing ring insert **524** can be made of a suture permeable material for suturing the prosthetic valve to a native annulus, as known in the art. For example, the sewing ring insert **524** can be made of a silicone-based material, although other suture-permeable materials can be used. The cloth cover **526** can be formed of any biocompatible fabric, such as, for example, polyethylene terephthalate or polyester fabric.

[0185] A skirt **528** can completely cover the frame **510**, including the base portion **512** and the commissure posts **518**. The sewing ring **522** can be secured to the frame **510** by being stitched to the skirt **528**, or via sutures extending through the sewing ring **522**, and optionally apertures of the base portion **512**, when the base portion **512** comprises such apertures. The skirt **528** can be formed of any biocompatible fabric, such as, for example, polyethylene terephthalate or polyester fabric.

[0186] FIG. **26** illustrates one example of a conventional surgically implantable prosthetic valve **500**.sup.a. Prosthetic valve **500**.sup.a is an exemplary implementation of prosthetic valve **500**, and thus includes all of the features described for prosthetic valve **500** above, except that the leaflets **562** are coupled to the base portion **512**.sup.a of the frame **510**.sup.a by sutures. In some configurations, as illustrated, the surgically implantable prosthetic valve **500**.sup.a further comprises an undulating wireform **520**, configured to provide further support to the leaflets **562**. The wireform **520** can include a plurality (e.g., three) large radius wireform cusps supporting the cusp regions of the valvular structure **560**, while the ends of each pair of adjacent wireform cusps converge somewhat asymptotically to form upstanding wireform commissure portions that terminate in tips, each extending in the opposite direction as the arcuate wireform cusps and having a relatively smaller radius. The cusp portions **516**.sup.a and the commissure posts **518**.sup.a can be sized and shaped so as to correspond to the curvature of the wireform **520**.

[0187] Wireform **520** typically is formed from one or more pieces of wire but also can be formed from other similarly-shaped elongate members. The wireform can also be cut or otherwise formed from tubing or a sheet of material. The wireform can have any of various cross sectional shapes, such as a square, rectangular, circular, or combinations thereof. In some implementations, wireform **520** is made of a relatively rigid metal, such as stainless steel or Elgiloy (a Co—Cr—Ni alloy). In some implementations, the wireform **520** further comprises a wireform cloth encapsulating it along its length.

[0188] In some implementations, the base portion **512**.sup.a can be in the form of a band, the leaflets **562** can be sutured along cusp edges **564** thereof to corresponding cusp portion **516**.sup.a

of the base portion **512**.sup.a and extend up along adjacent commissure posts **518**.sup.a. Each pair of aligned tabs **568** can be inserted between adjacent upstanding wireform portions. The tabs **568** can then be wrapped around a respective commissure posts **518**.sup.a of the frame **510**.sup.a, and sutured to each other and/or to the commissure post **518**.sup.a to form commissures **580**. The wireform **520** and the commissure posts **518**.sup.a can provide flexibility to the commissures **580** which helps reduce stress on the bioprosthetic material of the leaflets **562**. The wireform **520** is secured to the inner side of the frame **510**.sup.a, wherein the skirt **528** can, in some examples, cover the wireform **520** as well.
[0189] FIGS. **27-29**B show an exemplary surgically implantable prosthetic valve **500**.sup.b. Prosthetic valve **500**.sup.b is an exemplary implementation of prosthetic valve **500**, and thus includes all of the features described for prosthetic valve **500** above, except that the leaflets **562***b* are coupled to the base portion **512***b* of the frame **510**.sup.b by one or more gripping tubes **400**. FIG. **27** is an exploded view of components of prosthetic valve **500**.sup.b, including valvular structure **560**, frame **510**.sup.b and gripping tubes **400**, prior to attachment to each other. In some implementations, the frame **510**.sup.b can be formed as a wire or tubular member, having a circular or elliptical cross-sectional shape, though other cross-sectional shapes of the frame **510**.sup.b are contemplated. When formed with a circular cross-sectional shape, the frame **510**.sup.b, or at least cusp portion **516**.sup.b thereof, can define a frame tubular diameter D.sub.W. If the frame **510**.sup.b is not circular, its width, or the width of cusp portion **516**.sup.b thereof, can be equal to D.sub.W.
[0190] FIG. **28** shows a portion of assembled prosthetic valve **500**.sup.b, with a gripping tube **400** clamped around a cusp portion **516**.sup.b with a portion of leaflet **562** wrapped around the cusp portion **516**.sup.b inside an inner channel **404** defined by the gripping tube **400**. Gripping tube **400** is configured to couple a leaflet **562** to a cusp portion **516**.sup.b of frame **510**.sup.b by clamping over a portion of the leaflet **562**, for example along the cusp edge **564**, wrapped around a corresponding cusp portion **516**.sup.b, such that the gripping tube **400** snap-fits or clips over the wrapped cusp portion **516**.sup.b.
[0191] Gripping tube **400** comprises a mid-portion **406** and a couple of side arms **408** biased towards each other, the side arms **408** extending in a continuous manner from both sides of the mid-portion **406** and terminating at free ends **410**. As illustrated, the gripping tube **400** includes a slot **402**, and can be C-shaped in cross-section with the side arms **408** arched inward and terminating at free ends **410** on both sides of the slot **402**, such that the slot **402** defines a gap between the free ends **410**, through which a portion of the frame **510**.sup.b, such as cusp portion **516**.sup.b or any other part of the base portion **512***b*, optionally wrapped by a portion of leaflet **562**, may pass toward the mid-portion **406**. In some implementation, the gripping tube **400** if formed as a single monolithic component, for example formed from a single tube, along which a slit is cut to form the slot **402**. In alternative implementations, the gripping tube **400** may be comprised of more than one component, for example by having two separately formed side arms that may be affixed to two opposing edges of a mid-portion. The slot **402** defines a slot width S.sub.G in a free state of the gripping tube **400**. In some examples, the cross-sectional shape of the inner channel **404** of gripping tube **400** is similar, yet greater in size, than the cross-sectional shape of frame **510**.sup.b and/or cusp portions **516**.sup.b thereof.
[0192] The side arms **408** are resiliently expandable away from each other, such that a cusp portion **516**.sup.b, optionally wrapped, fully or partially, by leaflet **562**, is passable through the slot **402** formed between their free ends **410**. In use, a gripping tube **400** can be pushed over a cusp portion **516**.sup.b which is wrapped fully or partially by leaflet **562**. The slot width S.sub.G can be smaller than the frame tubular diameter D.sub.W. When the gripping tube **400** is pushed over the wrapped a cusp portion **516**.sup.b, the cusp portion **516**.sup.b, optionally with the portion of the leaflet **562** wrapped therearound, may apply a force sufficient to expand the side arms **408** away from each other, so as to allow the wrapped a cusp portion **516**.sup.b to pass through the slot **402** into inner

channel **404**, toward the mid-portion **406**. Once the cusp portion **516**.sup.b is completely accommodated within inner channel **404**, and in the absence of further expanding force applied to the side arms **408**, the side arms resiliently snap back toward each other to compress against the portion of the leaflet **562** extending from the cusp portion **516**.sup.b through the slot **402**, in order to clamp the leaflet **562** around the cusp portion **516**.sup.b of frame **510**.sup.b.

[0193] FIG. **29**A shows a cross-sectional view of an optional configuration in which the leaflet **562** is partially wrapped around the frame **510**.sup.b, such as around a cusp portion **516**.sup.b. In this configuration, the leaflet **562** is wrapped around a part of the circumference of frame **510**.sup.b, with the cusp edge **564** positioned within the inner channel **404**, such that only a single layer of leaflet **562** extends through slot **402**. In some examples, the gap width S.sub.G, in a free state of the gripping tube **400**, is smaller than the leaflet thickness T.sub.L, such that the side arms **408** snap towards each other against the portion of the leaflet **562** extending through the slot **402**. In some cases, the leaflet material, such as pericardium, is squeezable, for example up to half of its free-state thickness, such that the side arms **408** can pinch the leaflet **562** and optionally retain it in a somewhat squeezed thickness within the slot **402**, with both of the free end **410** pressing against both sides of the leaflet.

[0194] FIG. **29**B shows a cross-sectional view of an optional configuration in which the leaflet **562** is partially wrapped around the frame **510**.sup.b, such as around a cusp portion **516**.sup.b. In this configuration, the leaflet **562** extends into the inner channel **404** through the gap of slot **402**, circumscribes the frame **510**.sup.b, and extends back through the slot **402** out of the inner channel **404**, such that the cusp edge **564** is disposed out of the inner channel **404**, resulting in two layers of the leaflet **562** extending through slot **402**. The side arms **408** in this configurations can snap towards each other against the portions of the leaflet **562** extending into and out of the slot **402**. In some examples, the slot width S.sub.G, in a free state of the gripping tube **400**, is smaller than twice the leaflet thickness T.sub.L (i.e., S.sub.G<2T.sub.L), such that the side arms **408** can pinch both layers of the leaflet **562** extending through the slot **402** against each other, with one of the free ends **410** pressing against one layer of the leaflet **562** extending through the slot **402** into the inner channel, and the other free end **410** pressing against the layer of the leaflet **562** extending through the slot **402** out of the inner channel **404**.

[0195] The gripping tube **400** defines an inner channel diameter D.sub.IC. In some examples, the inner channel diameter D.sub.IC is larger than the frame tubular diameter D.sub.W. In some examples, the inner channel diameter D.sub.IC is larger than the sum of the frame tubular diameter D.sub.W and the thickness of one layer of the leaflet (i.e., D.sub.IC>(D.sub.W+T.sub.L)). In some examples, the inner channel diameter D.sub.IC is equal to or smaller than the sum of the frame tubular diameter D.sub.W and twice the thickness the leaflet (i.e., D.sub.IC≤(D.sub.W+2T.sub.L)). In some examples, the inner channel diameter D.sub.IC is smaller than the sum of the frame tubular diameter D.sub.W and three times the thickness the leaflet (i.e., D.sub.IC< (D.sub.W+3T.sub.L)).

[0196] While not explicitly shown for prosthetic valve **500**.sup.b, it is to be understood that additional components, such as a sewing ring **522** and/or a skirt **528**, cab be added after coupling the valvular structure **560** to the frame **510**.sup.b, in a similar manner to that described above for surgically implantable prosthetic valve **500**.

[0197] In some examples, a plurality of gripping tubes **400** are used to couple a valvular structure **560** to the frame **510**.sup.b. FIG. **27** shows a plurality of exemplary gripping tubes **400**.sup.a, and more specifically, three gripping tubes **400**.sup.a, corresponding to the three cusp portions **516**.sup.b of the frame **510**.sup.b. Each gripping tube **400**.sup.a can have a curved shape generally mimicking the curved shape of the corresponding cusp portion **516**.sup.b, and having a length that extends along whole or part of the length of the corresponding cusp portion **516**.sup.b. While the number of gripping tubes **400**.sup.a illustrated in FIG. **27** matches the number of cusp portion **516**.sup.b, it is to be understood that in some examples, a plurality of relatively shorter gripping tubes **400** can be clamped over a single corresponding cusp portion **516**.sup.b.

[0198] In some examples, one or more gripping tubes **400** can be similarly utilized to couple leaflets **562** not only to the cusp portion **516**.sup.b, but rather to the vertically oriented commissure posts **518**.sup.b of frame **510**.sup.b as well. FIG. **30** shows an exemplary gripping tube **400**.sup.b, which can be a single unit utilized to clamp the entire valvular structure **560** to frame **510**.sup.b. The shape of the gripping tube **400**.sup.b can generally mimic the shape of the entire frame **510**.sup.b. The gripping tube **400**.sup.b can include gripping tube cusp portions **412**, following the shape of the cusp portions **516**.sup.b of the frame **510**.sup.b, and gripping tube commissure portions **414**, corresponding to the U-shaped vertically oriented commissure posts **518**.sup.b of the frame **510**.sup.b. In such cases, the leaflet **562** may be provided without tabs **568**, but rather shaped to include a continuous edge that can be wrapped around the entire frame **510**.sup.b, and clamped thereto by the gripping tube **400**.sup.b.

[0199] In some examples, a plurality of differently shaped gripping tubes **400** can be utilized to clamp the valvular structure **560** to different portions of the frame **510**.sup.b. For example, a plurality (e.g., three) gripping tubes **400**, shaped in a similar manner to gripping tubes **400**.sup.a, can be used to couple the leaflets **562** to the cusp portions **516**.sup.b of the frame **510**.sup.b, and a separate plurality (e.g., three) of gripping tubes **400**, which can be shaped in a similar manner to U-shaped gripping tube commissure portions **414** of gripping tube **400**.sup.b, can be utilized to couple the leaflets **562** to the vertically oriented commissure posts **518**.sup.b of the frame **510**.sup.b.

Some Examples of the Disclosed Implementations

[0200] Some examples of above-described implementations are enumerated below. It should be noted that one feature of an example in isolation or more than one feature of the example taken in combination and, optionally, in combination with one or more features of one or more examples below are examples also falling within the disclosure of this application.

[0201] Example 1. A prosthetic valve comprising: [0202] a frame movable between a radially compressed state and a radially expanded state; [0203] a valvular structure coupled to the frame and comprising a plurality of leaflets configured to regulate flow through the prosthetic valve; and [0204] at least one stapling member coupling at least one of the plurality of leaflets to the frame, the at least one stapling member comprising: [0205] a spine extending along a stapling member axis; and [0206] a plurality of stapling portions separated from each other by intermediate portions of the spine, wherein each stapling portion comprises at least one leg extending from a backspan; [0207] wherein the at least one leg of each stapling portion extends through a corresponding one of the plurality of leaflets, and is bent around a corresponding strut of a plurality of struts of the frame, so as to couple the leaflet to the frame.

[0208] Example 2. The prosthetic valve of any example herein, particularly example 1, wherein each leaflet comprises: [0209] a cusp edge; [0210] a free edge opposite to the cusp edge; [0211] a cusp end portion extending from the cusp edge; and [0212] a leaflet body extending from the cusp edge to the free edge; [0213] wherein the at least one leg extending through each of the plurality of leaflets, extends through the cusp end portion of the leaflet.

[0214] Example 3. The prosthetic valve of any example herein, particularly example 1 or 2, wherein the backspan is perpendicular to the stapling member axis.

[0215] Example 4. The prosthetic valve of any example herein, particularly example 1 to 3, wherein the at least one leg of each stapling portion is angled relative to the stapling member axis.

[0216] Example 5. The prosthetic valve of any example herein, particularly example 4, wherein the at least one leg of each stapling portion is orthogonal to the stapling member axis.

[0217] Example 6. The prosthetic valve of any example herein, particularly any one of examples 1 to 5, wherein the at least one leg terminates at a tip configured to pierce through the corresponding leaflet.

[0218] Example 7. The prosthetic valve of any example herein, particularly example 6, wherein the tip comprises a slanted surface.

[0219] Example 8. The prosthetic valve of any example herein, particularly example 6, wherein the

tip is arrow-shaped.
[0220] Example 9. The prosthetic valve of any example herein, particularly example 6, wherein the tip is conical.
[0221] Example 10. The prosthetic valve of any example herein, particularly any one of examples 1 to 5, wherein each leaflet comprises at least one pre-formed aperture through which a corresponding one of the at least one legs extends.
[0222] Example 11. The prosthetic valve of any example herein, particularly example 10, wherein the at least one leg terminates at a blunt tip.
[0223] Example 12. The prosthetic valve of any example herein, particularly any one of examples 1 to 11, wherein the leaflet is sandwiched between the spine and the frame.
[0224] Example 13. The prosthetic valve of any example herein, particularly example 2, wherein the cusp end portion is pressed between the backspan and the strut.
[0225] Example 14. The prosthetic valve of any example herein, particularly any one of examples 1 to 13, wherein the spine defines a flat spine outer surface which is pressed against the leaflet coupled thereby to the frame.
[0226] Example 15. The prosthetic valve of any example herein, particularly any one of examples 1 to 14, wherein the spine and the stapling portions are integrally formed.
[0227] Example 16. The prosthetic valve of any example herein, particularly any one of examples 1 to 14, wherein the stapling portion comprises a stapling element provided as a separately formed component which is attached to the spine, wherein the stapling element comprises the backspan and the at least one leg.
[0228] Example 17. The prosthetic valve of any example herein, particularly example 16, wherein the backspan is attached to the spine.
[0229] Example 18. The prosthetic valve of any example herein, particularly any one of examples 1 to 17, wherein each leg of the stapling portion is bent at a first bend relative to the backspan.
[0230] Example 19. The prosthetic valve of any example herein, particularly example 18, wherein the first bend defines a right angle between the backspan and the corresponding leg.
[0231] Example 20. The prosthetic valve of any example herein, particularly example 18 or 19, wherein the at least one leg is bent at a second bend that divides the leg to a radial portion extending between the first bend and the second bend, and a clinch portion extending from the second bend.
[0232] Example 21. The prosthetic valve of any example herein, particularly example 20, wherein the second bend defines a right angle between the radial portion and the clinch portion.
[0233] Example 22. The prosthetic valve of any example herein, particularly any one of examples 20 to 21, wherein the radial portion of each stapling portion extends radially through the corresponding leaflet and across the corresponding strut the leaflet is coupled to.
[0234] Example 23. The prosthetic valve of any example herein, particularly any one of examples 20 to 22, wherein the radial portion defines a radial portion length which is at least as great as the sum of thicknesses of the leaflet and the strut.
[0235] Example 24. The prosthetic valve of any example herein, particularly any one of examples 20 to 23, wherein the at least one leg is bent at a third bend that divides the clinch portion to an axial clinch section extending between the second bend and the third bend, and a radial clinch section extending from the third bend.
[0236] Example 25. The prosthetic valve of any example herein, particularly example 24, wherein the radial clinch section is parallel to the radial portion.
[0237] Example 26 The prosthetic valve of any example herein, particularly example 24 or 25, wherein the leg defines a leg length, wherein the strut defines a strut thickness and a strut width, and wherein the leg length is greater than the sum of the strut width and the strut thickness.
[0238] Example 27. The prosthetic valve of any example herein, particularly example 26, wherein the leg length is smaller than the sum of the strut width and twice the strut thickness.

[0239] Example 28. The prosthetic valve of any example herein, particularly any one of examples 1 to 27, wherein the at least one leg of the stapling portion comprises a single leg.
[0240] Example 29. The prosthetic valve of any example herein, particularly any one of examples 6 to 9 or 11, wherein the at least one leg of the stapling portion comprises a pair of legs, and wherein the tips of the pair of legs are oriented towards each other.
[0241] Example 30. The prosthetic valve of any example herein, particularly any one of examples 18 to 23, wherein the at least one leg of the stapling portion comprises a pair of legs extending from opposite ends of the backspan.
[0242] Example 31. The prosthetic valve of any example herein, particularly example 30, and wherein the clinch portions of the legs of the stapling portion are oriented towards each other.
[0243] Example 32. The prosthetic valve of any example herein, particularly any one of examples 30 to 31, wherein the legs of the stapling portion are aligned with each other.
[0244] Example 33. The prosthetic valve of any example herein, particularly example 33, wherein the clinch portions of the stapling portion are coaxial.
[0245] Example 34. The prosthetic valve of any example herein, particularly example 32 or 33, wherein each clinch portion defines a clinch portion length, and wherein the backspan defines a backspan width, such that the backspan width is at least two times greater the clinch portion length.
[0246] Example 35. The prosthetic valve of any example herein, particularly example 32 or 33, wherein each clinch portion defines a clinch portion length, and wherein the strut defines a strut width, such that the strut width is at least two times greater the clinch portion length.
[0247] Example 36. The prosthetic valve of any example herein, particularly example 30, wherein the legs of the stapling portion are offset from each other.
[0248] Example 37. The prosthetic valve of any example herein, particularly example 36, wherein the clinch portions of the stapling portion are parallel to each other.
[0249] Example 38. The prosthetic valve of any example herein, particularly example 36 or 37, wherein each clinch portion defines a clinch portion length, and wherein the backspan defines a backspan width, such that the clinch portion length is greater than half the backspan width.
[0250] Example 39. The prosthetic valve of any example herein, particularly example 36 or 37, wherein each clinch portion defines a clinch portion length, and wherein the strut defines a strut width, such that the clinch portion length is greater than half the strut width.
[0251] Example 40. The prosthetic valve of any example herein, particularly any one of examples 29 to 39, wherein one leg of the pair of legs is shorter than the other leg of the pair of legs.
[0252] Example 41. The prosthetic valve of any example herein, particularly any one of examples 20 to 40, wherein the clinch portion extends over an outer surface of the strut.
[0253] Example 42. The prosthetic valve of any example herein, particularly any one of examples 1 to 41, wherein the stapling member is formed from a plate cut to form a planar configuration of the stapling member prior to being used for coupling the leaflet to the frame, and bent to form a second bent configuration of the stapling member when used to couple the leaflet to the frame.
[0254] Example 43. The prosthetic valve of any example herein, particularly any one of examples 1 to 41, wherein the stapling member is formed from a wire, and wherein the intermediate portions are intermediate wire portions.
[0255] Example 44. The prosthetic valve of any example herein, particularly example 43, [0256] wherein the wire is bent to form the plurality of stapling portions in a planar configuration of the stapling member prior to being used for coupling the leaflet to the frame, wherein each stapling portion comprises, in the planar configuration of the stapling member: [0257] a first planar bent extension, terminating at a first tip; [0258] a second planar bent extension, extending from the first tip to an opposite second tip; and [0259] a third planar bent extension extending from the second tip.
[0260] Example 45. The prosthetic valve of any example herein, particularly example 44, wherein the first tip comprises a first lateral bent extension between the first planar bent extension and the

second planar bent extension, and wherein the second tip comprises a second lateral bent extension between the second planar bent extension and the third planar bent extension.

[0261] Example 46. The prosthetic valve of any example herein, particularly example 44, wherein the first tip defines an acute angle between the first planar bent extension and the second planar bent extension, and wherein the second tip defines a sharp angle between the second planar bent extension and the third planar bent extension.

[0262] Example 47. The prosthetic valve of any example herein, particularly any one of examples 1 to 46, wherein each cusp end portion comprises a stiff portion which is stiffer than a leaflet body of the leaflet.

[0263] Example 48. The prosthetic valve of any example herein, particularly example 47, wherein the stiff portion defines a stiff portion width between a cusp edge of the leaflet and a stiff portion proximal end, such that a leaflet body extends between the stiff portion proximal end and a free edge of the leaflet.

[0264] Example 49. The prosthetic valve of any example herein, particularly example 48, wherein the stiff portion width is at least 1.5 times greater than a strut width defined by the strut to which the leaflet is coupled by the stapling member.

[0265] Example 50. The prosthetic valve of any example herein, particularly example 49, wherein the stiff portion width is at least 2 times greater than a strut width.

[0266] Example 51. The prosthetic valve of any example herein, particularly any one of examples 48 to 50, wherein the stiff portion has an ultimate tensile stress which is at least 1.5 times greater than an ultimate tensile stress of the leaflet body.

[0267] Example 52. The prosthetic valve of any example herein, particularly example 51, wherein the ultimate tensile stress of the stiff portion is at least 2 times greater than the ultimate tensile stress of the leaflet body.

[0268] Example 53. The prosthetic valve of any example herein, particularly any one of examples 48 to 52, wherein the stiff portion obtains a load at failure which is at least 2 times greater than a load at failure obtained by the leaflet body.

[0269] Example 54. The prosthetic valve of any example herein, particularly any one of examples 48 to 53, wherein the stiff portion is coated by a reinforcing layer.

[0270] Example 55. The prosthetic valve of any example herein, particularly any one of examples 1 to 54, wherein the stapling member axis is curved.

[0271] Example 56. The prosthetic valve of any example herein, particularly any one of examples 1 to 55, wherein the at least one stapling member comprises a plurality of stapling members, together forming a scalloped-shaped line along which the valvular structure is coupled to the frame by the stapling members.

[0272] Example 57. The prosthetic valve of any example herein, particularly any one of examples 1 to 54, wherein the spine is zig-zagged.

[0273] Example 58. The prosthetic valve of any example herein, particularly any one of examples 1 to 54, wherein the spine comprises at least one offsetting portions.

[0274] Example 59. The prosthetic valve of any example herein, particularly example 58, wherein the at least one offsetting portion is aligned with at least one junction of the frame.

[0275] Example 60. The prosthetic valve of any example herein, particularly any one of examples 1 to 59, wherein the plurality of leaflets comprises three leaflets.

[0276] Example 61. The prosthetic valve of any example herein, particularly any one of examples 1 to 60, wherein the stapling member is formed of a plastically deformable material.

[0277] Example 62. The prosthetic valve of any example herein, particularly any one of examples 1 to 61, further comprising an inner skirt coupled to the frame.

[0278] Example 63. The prosthetic valve of any example herein, particularly example 62, wherein the legs extend through the inner skirt.

[0279] Example 64. The prosthetic valve of any example herein, particularly any one of examples 1

to 63, further comprising an inner skirt coupled to the frame.
[0280] Example 65. A method of assembling a prosthetic valve, comprising: [0281] providing a stapling member in a first bent configuration, comprising a spine extending along a stapling member axis and a plurality of U-shaped stapling portions separated from each other by intermediate portions of the spine, each stapling portion comprising a backspan and a pair of legs bent relative to the backspan at first bends of the legs; [0282] approximating a leaflet to a strut of a frame of a prosthetic valve; [0283] pushing the legs toward the leaflet such that tips of the legs pierce through the leaflet; [0284] extending the legs radially through the leaflet and across the strut; and [0285] transitioning the stapling member to a second bent configuration by bending the legs at second bends thereof over the struts, so as to couple the leaflet to the strut.
[0286] Example 66. The method of any example herein, particularly example 65, wherein the legs are parallel to each other at the first bent configuration.
[0287] Example 67. The method of any example herein, particularly example 65 or 66, wherein approximating a leaflet to the strut comprises approximating a cusp end portion of the leaflet to the strut.
[0288] Example 68. The method of any example herein, particularly example 67, wherein pushing the legs toward the leaflet comprising pushing the legs toward the cusp end portion such that tips of the legs pierce through the cusp end portion.
[0289] Example 69. The method of any example herein, particularly example 67 or 68, wherein extending the legs radially through the leaflet comprising extending the legs radially through the cusp end portion.
[0290] Example 70. The method of any example herein, particularly example 65 or 66, wherein approximating a leaflet to the strut comprises approximating a stiff portion of the leaflet to the strut, and wherein the leaflet comprises a leaflet body extending between the stiff portion and a free edge of the leaflet.
[0291] Example 71. The method of any example herein, particularly example 70, wherein pushing the legs toward the leaflet comprising pushing the legs toward the stiff portion such that tips of the legs pierce through the stiff portion.
[0292] Example 72. The method of any example herein, particularly example 70 or 71, wherein extending the legs radially through the leaflet comprising extending the legs radially through the stiff portion.
[0293] Example 73. The method of any example herein, particularly any one of examples 70 to 72, wherein the stiff portion has an ultimate tensile stress which is at least 1.5 times greater than an ultimate tensile stress of the leaflet body.
[0294] Example 74. The method of any example herein, particularly example 73, wherein the ultimate tensile stress of the stiff portion is at least 2 times greater than the ultimate tensile stress of the leaflet body.
[0295] Example 75. The method of any example herein, particularly any one of examples 70 to 74, wherein the stiff portion obtains a load at failure which is at least 2 times greater than a load at failure obtained by the leaflet body.
[0296] Example 76. The method of any example herein, particularly any one of examples 70 to 75, wherein the stiff portion is coated by a reinforcing layer.
[0297] Example 77. The method of any example herein, particularly any one of examples 65 to 76, wherein extending the legs radially through the leaflet comprising sandwiching the leaflet between the backspan and the strut.
[0298] Example 78. The method of any example herein, particularly any one of examples 65 to 77, wherein each tip comprises a slanted surface.
[0299] Example 79. The method of any example herein, particularly any one of examples 65 to 77, wherein each tip is arrow-shaped.
[0300] Example 80. The method of any example herein, particularly any one of examples 65 to 77,

wherein each tip is conical.
[0301] Example 81. The method of any example herein, particularly any one of examples 65 to 78, wherein bending the legs comprises orienting the tips of each stapling member toward each other.
[0302] Example 82. The method of any example herein, particularly any one of examples 65 to 78, wherein pushing the legs toward the leaflet comprises pushing the legs toward an inner surface of the leaflet.
[0303] Example 83. The method of any example herein, particularly any one of examples 65 to 82, wherein bending the legs over the struts comprises bending the legs over an outer surface of the strut.
[0304] Example 84. The method of any example herein, particularly any one of examples 65 to 83, wherein each leg comprises, in the second bent configuration, a radial portion extending from the first bend to the second bend along a radial portion length, and a clinch portion extending between the second bend and the tip along a clinch portion length.
[0305] Example 85. The method of any example herein, particularly example 84, wherein the radial portion length is at least as great as the combined thicknesses of the leaflet and the strut.
[0306] Example 86. The method of any example herein, particularly example 84 or 85, wherein the radial portions are parallel to each other.
[0307] Example 87. The method of any example herein, particularly any one of examples 84 to 86, wherein the legs of each stapling portion are aligned with each other.
[0308] Example 88. The method of any example herein, particularly example 87, wherein the clinch portions of the stapling portion are coaxial.
[0309] Example 89. The method of any example herein, particularly example 87 or 88, wherein the backspan defines a backspan width, such that the backspan width is at least two times greater the clinch portion length.
[0310] Example 90. The method of any example herein, particularly example 87 or 88, wherein the strut defines a strut width, and wherein the strut width is at least two times greater the clinch portion length.
[0311] Example 91. The method of any example herein, particularly any one of examples 84 to 86, wherein the legs of the stapling portion are offset from each other.
[0312] Example 92. The method of any example herein, particularly example 91, wherein the clinch portions of the stapling portion are parallel to each other.
[0313] Example 93. The method of any example herein, particularly example 91 or 92, the backspan defines a backspan width, such that the clinch portion length is greater than half the backspan width.
[0314] Example 94. The method of any example herein, particularly example 91 or 92, wherein the strut defines a strut width, and wherein the clinch portion length is greater than half the strut width.
[0315] Example 95. The method of any example herein, particularly any one of examples 65 to 94, wherein providing the stapling member in a first bent configuration comprises: [0316] providing the stapling member in a flattened configuration; and [0317] transitioning the stapling member to the first bent configuration by bending the legs at the first bends.
[0318] Example 96. The method of any example herein, particularly example 95, wherein bending the legs at the first bends comprises bending the legs at right angles.
[0319] Example 97. The method of any example herein, particularly example 95 or 97, wherein the legs of each stapling member are coplanar with the intermediate portion and the spine in the flattened configuration.
[0320] Example 98. The method of any example herein, particularly any one of examples 95 to 97, wherein providing the stapling member in a flattened configuration comprises cutting the stapling member from a plate.
[0321] Example 99. The method of any example herein, particularly any one of examples 95 to 97, wherein providing the stapling member in a flattened configuration comprises bending a wire from

which the stapling member is made to form the flattened configuration.
[0322] Example 100. The method of any example herein, particularly example 99, wherein bending the wire to form the flattened configuration comprises bending the wire to form each of the stapling portions.
[0323] Example 101. The method of any example herein, particularly example 100, wherein bending the wire to form each stapling portion comprises: [0324] laterally bending the wire relative to the stapling member axis, forming a first planar bent extension terminating at a first tip; [0325] extending a second planar bent extension of the wire from the first tip toward a second tip, wherein the first tip and the second tip are positioned at opposite sides of the stapling member axis; and [0326] extending a third planar bent extension from the second tip toward the stapling member axis.
[0327] Example 102. The method of any example herein, particularly example 101, further comprising, after forming a first planar bent extension and prior to extending the second planar bent extension, bending the wire relative to the first planar bent extension to form a first lateral bent extension, and bending the first lateral bent extension to form the second planar bent extension extending therefrom.
[0328] Example 103. The method of any example herein, particularly example 102, further comprising, after extending the second planar bent extension and prior to extending the third planar bent extension, bending the wire relative to the second planar bent extension to form a second lateral bent extension, and bending the second lateral bent extension to form the third planar bent extension extending therefrom.
[0329] Example 104. The method of any example herein, particularly example 101, wherein extending a second planar bent extension comprises bending the wire to as to form an acute angle between the first planar bent extension and the second planar bent extension.
[0330] Example 105. The method of any example herein, particularly example 104, wherein extending a third planar bent extension comprises bending the wire to as to form an acute angle between the second planar bent extension and the third planar bent extension.
[0331] Example 106. The method of any example herein, particularly any one of examples 65 to 105, wherein the stapling member axis is curved.
[0332] Example 107. The method of any example herein, particularly any one of examples 65 to 106, wherein the stapling member is formed of a plastically deformable material.
[0333] Example 108. A prosthetic valve comprising: [0334] a frame movable between a radially compressed state and a radially expanded state, the frame comprising a plurality of intersecting struts, the plurality of intersecting struts comprising recessed struts and unrecessed struts; [0335] a valvular structure coupled to the frame and comprising a plurality of leaflets configured to regulate flow through the prosthetic valve; and [0336] at least one compressible insert disposed within a recess formed in at least one of the recessed struts; [0337] wherein at least one of the plurality of leaflets is coupled to at least one of the recessed struts by having a cusp end portion thereof clamped inside the recess of the corresponding recessed strut by the corresponding at least one compressible insert.
[0338] Example 109. The prosthetic valve of any example herein, particularly example 108, wherein the recess comprises an opening formed at an inner surface of the recessed strut, the opening defining an opening width.
[0339] Example 110. The prosthetic valve of any example herein, particularly example 109, wherein the recess defines a maximal recess width in a direction parallel to the opening width, and wherein the opening width is smaller than the recess width.
[0340] Example 111. The prosthetic valve of any example herein, particularly example 110, wherein the compressible insert defines an insert diameter in a free state thereof, which is compressible when pushed through the recess opening, and strives to resiliently return to the insert diameter of its free state when positioned within the recess.

[0341] Example 112. The prosthetic valve of any example herein, particularly example 111, wherein the insert diameter in the free state is greater than the opening width.
[0342] Example 113. The prosthetic valve of any example herein, particularly example 111 or 112, wherein the insert diameter in the free state is greater than a gap defined between layers of the leaflet extending through the opening.
[0343] Example 114. The prosthetic valve of any example herein, particularly any one of examples 110 to 113, wherein the recessed strut defines a strut width which is greater than the maximal recess width.
[0344] Example 115. The prosthetic valve of any example herein, particularly example 114, wherein the unrecessed struts are devoid of recesses.
[0345] Example 116. The prosthetic valve of any example herein, particularly example 115, wherein a strut width defined by at least one of the unrecessed struts is smaller than the strut width of the recessed strut.
[0346] Example 117. The prosthetic valve of any example herein, particularly any one of examples 109 to 116, wherein the recess defines a recess depth, and wherein the recessed strut defines a strut thickness which is greater than the recess depth.
[0347] Example 118. The prosthetic valve of any example herein, particularly any one of examples 108 to 117, wherein the cusp end portion is at least partially wrapped around the compressible insert within the recess.
[0348] Example 119. The prosthetic valve of any example herein, particularly any one of examples 108 to 118, wherein the valvular structure is coupled, by the at least one compressible insert, to a plurality of recessed struts that collectively form a scalloped-shaped line.
[0349] Example 120. The prosthetic valve of any example herein, particularly example 119, wherein the at least one compressible insert comprises a plurality of compressible inserts, corresponding in number to the plurality of recesses formed in the plurality of recessed struts.
[0350] Example 121. A method of assembling a prosthetic valve, comprising: [0351] approximating a cusp end portion of at least one leaflet to an inner surface of a recessed strut of a frame of a prosthetic valve; and [0352] pushing a compressible insert, along with a portion of the cusp end portion, into a recess formed in the corresponding recessed strut, thereby coupling the cusp end portion to the frame.
[0353] Example 122. The method of any example herein, particularly example 121, wherein pushing a compressible insert comprises pushing the compressible insert through an opening of the recess, wherein the opening defines an opening width at the inner surface of the strut.
[0354] Example 123. The method of any example herein, particularly example 122, wherein the recess defines a maximal recess width in a direction parallel to the opening width, and wherein the opening width is smaller than the recess width.
[0355] Example 124. The method of any example herein, particularly example 121 or 122, wherein pushing the compressible insert comprises squeezing the compressible insert through the opening.
[0356] Example 125. The method of any example herein, particularly any one of examples 122 to 124, wherein the diameter of the compressible insert, in a free state thereof, is smaller than the opening width.
[0357] Example 126. The method of any example herein, particularly any one of examples 122 to 125, wherein the diameter of the compressible insert, when fully disposed within the recess, is greater than a gap formed between layers of the leaflet extending through the opening.
[0358] Example 127. The method of any example herein, particularly example 123, wherein the recessed strut defines a strut width which is greater than the maximal recess width.
[0359] Example 128. The method of any example herein, particularly example 127, wherein the frame further comprises a plurality of unrecessed struts which are devoid of recesses.
[0360] Example 129. The method of any example herein, particularly example 128, wherein a strut width defined by at least one of the unrecessed struts is smaller than the strut width of the recessed

strut.

[0361] Example 130. The method of any example herein, particularly any one of examples 121 to 129, wherein the recess defines a recess depth, and wherein the recessed strut defines a strut thickness which is greater than the recess depth.

[0362] Example 131. The method of any example herein, particularly any one of examples 121 to 130, wherein pushing a compressible insert comprises wrapping at least a portion of the cusp end portion around the compressible insert, such that the wrapped portion is sandwiched between the compressible insert and an inner wall of the recess.

[0363] Example 132. A prosthetic valve comprising: [0364] a frame comprising a base portion and a plurality of vertically oriented commissure posts extending from the base portion; [0365] a valvular structure coupled to the frame and comprising a plurality of leaflets configured to regulate flow through the prosthetic valve; and [0366] at least one gripping tube comprising: [0367] an inner channel; and [0368] a slot extending between free ends of side arms of the gripping tube; [0369] wherein at least one of the plurality of leaflets is coupled to the frame by the at least one gripping tube disposed around a cusp end portion of the leaflet which is at least partially wrapped around at least a part of the base portion retained within the inner channel.

[0370] Example 133. The prosthetic valve of any example herein, particularly example 132, wherein the slot defines a slot width in a free state of the slot, and wherein the side arms are configured to expand the slot when the gripping tube is pushed around the base portion, and strive to resiliently return towards each other to the slot width of the free state when no components disposed within the slot prevent them from doing so.

[0371] Example 134. The prosthetic valve of any example herein, particularly example 133, wherein the frame has a circular cross-section defining a frame tube diameter.

[0372] Example 135. The prosthetic valve of any example herein, particularly example 134, wherein the frame tube diameter is greater than the slot width.

[0373] Example 136. The prosthetic valve of any example herein, particularly any one of examples 133 to 135, wherein the leaflet is partially wrapped around the base portion of the frame, such that a cusp edge of the leaflet is disposed within the inner channel.

[0374] Example 137. The prosthetic valve of any example herein, particularly example 136, wherein the slot width is smaller than a thickness of the leaflet.

[0375] Example 138. The prosthetic valve of any example herein, particularly any one of examples 133 to 135, wherein the leaflet extends into the inner channel through the slot, is wrapped around the base portion of the frame inside the inner channel, and extends back out of the inner channel through the slot, such that a cusp edge of the leaflet is disposed out of the gripping tube.

[0376] Example 139. The prosthetic valve of any example herein, particularly example 138, wherein a thickness defined by the leaflet is greater than half the slot width.

[0377] Example 140. The prosthetic valve of any example herein, particularly any one of examples 132 to 139, wherein the base portion comprises a plurality of cusp portions disposed between the vertically oriented commissure posts, and wherein the leaflet is coupled, by the at least one gripping tube, to at least one of the cusp portions of the frame.

[0378] Example 141. The prosthetic valve of any example herein, particularly example 140, wherein the plurality of leaflets comprises three leaflets, wherein the plurality of cusp portions of the frame comprises three cusp portions, and wherein the plurality of vertically oriented commissure posts comprises three vertically oriented commissure posts.

[0379] Example 142. The prosthetic valve of any example herein, particularly example 141, wherein each leaflet of the three leaflets is coupled to a corresponding one of the three cusp portions.

[0380] Example 143. The prosthetic valve of any example herein, particularly example 142, wherein the at least one gripping tube comprises three gripping tubes, each of which coupling a corresponding one of the leaflets to a corresponding one of the cusp portions.

[0381] Example 144. The prosthetic valve of any example herein, particularly any one of examples 142 or 143, wherein each two of the three leaflets are further coupled to a corresponding one of the three vertically oriented commissure posts.
[0382] Example 145. The prosthetic valve of any example herein, particularly example 144, wherein the at least one gripping tube comprises a single gripping tube that comprises three gripping tube cusp portions, configured to couple the leaflets to the cusp portions of the frame, and three gripping tube commissure portions, configured to couple the leaflets to the vertically oriented commissure posts.
[0383] Example 146. The prosthetic valve of any example herein, particularly example 145, wherein the single gripping tube mimics the shape of the frame.
[0384] Example 147. The prosthetic valve of any example herein, particularly any one of examples 144 to 146, wherein each of the vertically oriented commissure posts is U-shaped.
[0385] Example 148. The prosthetic valve of any example herein, particularly any one of examples 132 to 147, further comprising a skirt coupled to the frame.
[0386] Example 149. The prosthetic valve of any example herein, particularly example 148, further comprising a sewing ring coupled to the skirt.
[0387] Example 150. A method of assembling a prosthetic valve, comprising: [0388] positioning a cusp end portion of at least one leaflet between a base portion of a frame of a prosthetic valve and at least one gripping tube; and [0389] pushing the gripping tube toward the cusp end portion and the base portion, such that a slot of the gripping tube expands to let the base portion and the cusp end portion slide into an inner channel of the gripping tube, until the base portion resides within the inner channel with the cusp end portion at least partially wrapped around the base portion, at which point side arms defining the slot of the gripping tube are allowed to snap back toward each and clamp against a portion of the leaflet extending through the slot, so as to couple the at least one leaflet to the base portion.
[0390] Example 151. The method of any example herein, particularly example 150, wherein the slot defines a slot width in a free state of the slot, prior to pushing the gripping tube.
[0391] Example 152. The method of any example herein, particularly example 151, wherein the frame has a circular cross-section defining a frame tube diameter.
[0392] Example 153. The method of any example herein, particularly example 152, wherein the frame tube diameter is greater than the slot width.
[0393] Example 154. The method of any example herein, particularly any one of examples 151 to 153, wherein pushing the gripping tube until the base portion resides within the inner channel with the cusp end portion at least partially wrapped around the base portion, comprises partially wrapping the cusp end portion around the base portion, such that a cusp edge of the leaflet is disposed within the inner channel.
[0394] Example 155. The method of any example herein, particularly example 154, wherein the slot width is smaller than a thickness of the leaflet.
[0395] Example 156. The method of any example herein, particularly any one of examples 151 to 153, wherein pushing the gripping tube until the base portion resides within the inner channel with the cusp end portion at least partially wrapped around the base portion, comprises extending the leaflet into the inner channel through the slot, wrapping the cusp end portions around the base portion inside the inner channel, and extending the leaflet back out of the inner channel through the slot, such that a cusp edge of the leaflet is disposed out of the gripping tube.
[0396] Example 157. The method of any example herein, particularly example 156, wherein a thickness defined by the leaflet is greater than half the slot width.
[0397] Example 158. The method of any example herein, particularly any one of examples 150 to 157, wherein the base portion comprises a plurality of cusp portions disposed between vertically oriented commissure posts of the frame.
[0398] Example 159. The method of any example herein, particularly example 158, wherein

pushing the gripping tube toward the base portion to couple the at least one leaflet to the base portion, comprises pushing the gripping tube towards at least one of the cusp portions to couple the at least one leaflet to the at least one of the cusp portions.
[0399] Example 160. The method of any example herein, particularly any one of examples 150 to 159, further comprising, after pushing the gripping tube to couple the at least one leaflet to the base portion, suturing a skirt around the frame so as to cover the frame by the skirt.
[0400] Example 161. The method of any example herein, particularly **160**, further comprising, after suturing the skirt, stitching a sewing ring to the skirt.
[0401] It is appreciated that certain features of the disclosure, which are, for clarity, described in the context of separate examples, may also be provided in combination in a single example. Conversely, various features of the disclosure, which are, for brevity, described in the context of a single example, may also be provided separately or in any suitable sub-combination or as suitable in any other described example of the disclosure. No feature described in the context of an example is to be considered an essential feature of that example, unless explicitly specified as such.
[0402] In view of the many possible examples to which the principles of the disclosure may be applied, it should be recognized that the illustrated examples are only preferred examples and should not be taken as limiting the scope. Rather, the scope is defined by the following claims. We therefore claim all that comes within the scope and spirit of these claims.

Claims

1. A prosthetic valve comprising: a frame movable between a radially compressed state and a radially expanded state; a valvular structure coupled to the frame and comprising a plurality of leaflets configured to regulate flow through the prosthetic valve; and at least one stapling member coupling at least one of the plurality of leaflets to the frame, the at least one stapling member comprising: a spine extending along a stapling member axis; and a plurality of stapling portions separated from each other by intermediate portions of the spine, wherein each stapling portion comprises at least one leg extending from a backspan; wherein the at least one leg of each stapling portion extends through a corresponding one of the plurality of leaflets, and is bent around a corresponding strut of a plurality of struts of the frame, so as to couple the leaflet to the frame.
2. The prosthetic valve of claim 1, wherein the leaflet is sandwiched between the spine and the frame.
3. The prosthetic valve of claim 1 wherein each leg of the stapling portion is bent at a first bend relative to the backspan.
4. The prosthetic valve of claim 3, wherein the at least one leg is bent at a second bend that divides the leg to a radial portion extending between the first bend and the second bend, and a clinch portion extending from the second bend.
5. The prosthetic valve of claim 1, wherein the at least one leg of the stapling portion comprises a single leg.
6. The prosthetic valve of claim 1, wherein the at least one leg of the stapling portion comprises a pair of legs, and wherein tips of the pair of legs are oriented towards each other.
7. The prosthetic valve of claim 6, wherein one leg of the pair of legs is shorter than the other leg of the pair of legs.
8. The prosthetic valve of claim 1, wherein the stapling member is formed from a plate cut to form a planar configuration of the stapling member prior to being used for coupling the leaflet to the frame, and bent to form a second bent configuration of the stapling member when used to couple the leaflet to the frame.
9. The prosthetic valve of claim 1, wherein the stapling member is formed from a wire, and wherein the intermediate portions are intermediate wire portions.
10. The prosthetic valve of claim 9, wherein the wire is bent to form the plurality of stapling

portions in a planar configuration of the stapling member prior to being used for coupling the leaflet to the frame, wherein each stapling portion comprises, in the planar configuration of the stapling member: a first planar bent extension, terminating at a first tip; a second planar bent extension, extending from the first tip to an opposite second tip; and a third planar bent extension extending from the second tip.

11. The prosthetic valve of claim 1, wherein each cusp end portion comprises a stiff portion which is stiffer than a leaflet body of the leaflet.

12. The prosthetic valve of claim 1, wherein the stapling member axis is curved.

13. The prosthetic valve of claim 1, wherein the spine is zig-zagged.

14. A method of assembling a prosthetic valve, comprising: providing a stapling member in a first bent configuration, comprising a spine extending along a stapling member axis and a plurality of U-shaped stapling portions separated from each other by intermediate portions of the spine, each stapling portion comprising a backspan and a pair of legs bent relative to the backspan at first bends of the legs; approximating a leaflet to a strut of a frame of a prosthetic valve; pushing the legs toward the leaflet such that tips of the legs pierce through the leaflet; extending the legs radially through the leaflet and across the strut; and transitioning the stapling member to a second bent configuration by bending the legs at second bends thereof over the struts, so as to couple the leaflet to the strut.

15. The method of claim 14, wherein the legs are parallel to each other at the first bent configuration.

16. The method of claim 14, wherein approximating a leaflet to the strut comprises approximating a stiff portion of the leaflet to the strut, and wherein the leaflet comprises a leaflet body extending between the stiff portion and a free edge of the leaflet.

17. The method of claim 14, wherein extending the legs radially through the leaflet comprising sandwiching the leaflet between the backspan and the strut.

18. The method of claim 14, wherein providing the stapling member in a first bent configuration comprises: providing the stapling member in a flattened configuration; and transitioning the stapling member to the first bent configuration by bending the legs at the first bends.

19. The method of claim 18, wherein providing the stapling member in a flattened configuration comprises cutting the stapling member from a plate.

20. The method of claim 18, wherein providing the stapling member in a flattened configuration comprises bending a wire from which the stapling member is made to form the flattened configuration.
